

Bitwise linear trail through **LBlock** found by SAT

CiVerLy

July 20, 2026

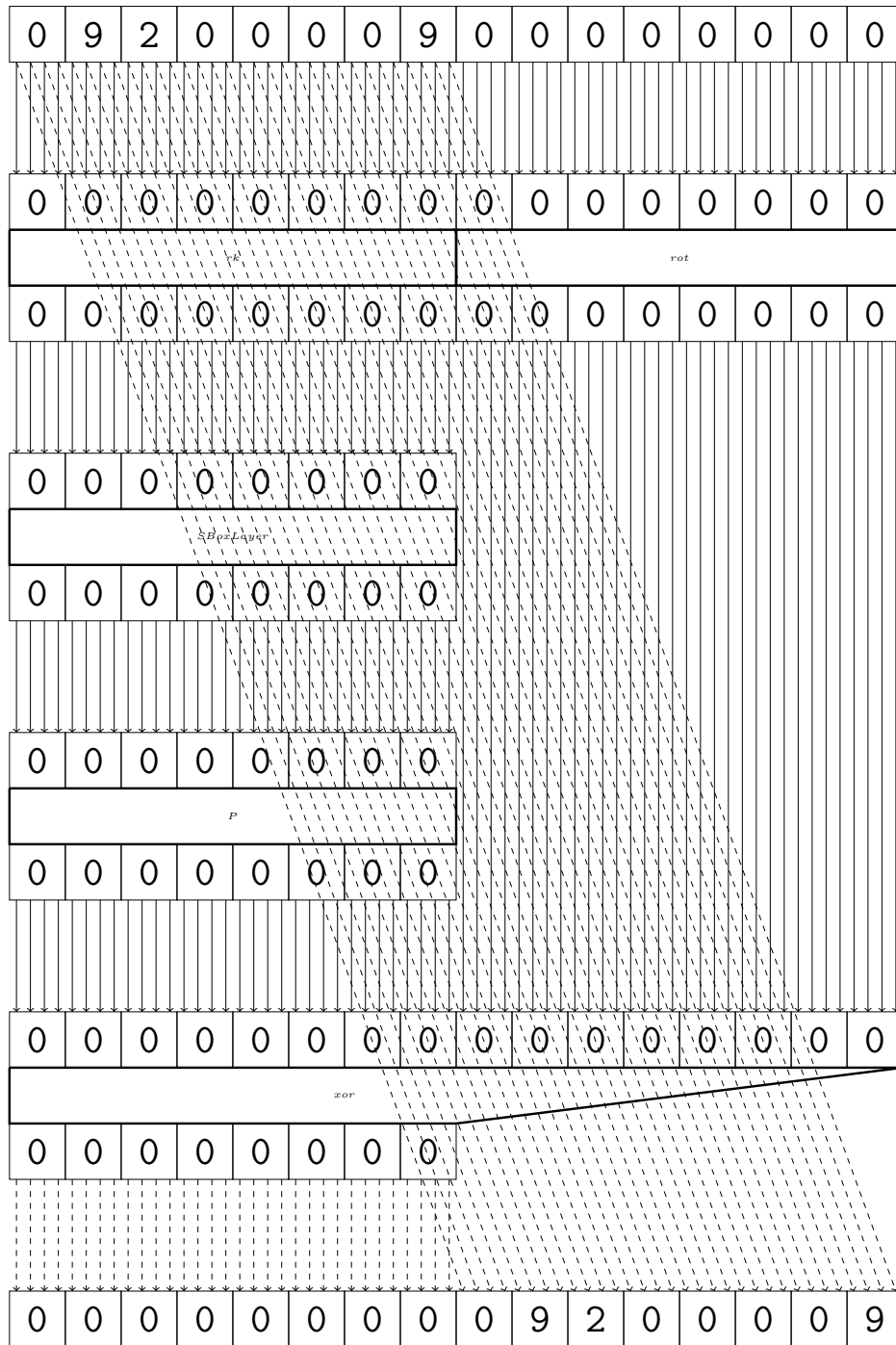
1 LBlock

Maximal linear correlation: 2^{-37}

0	9	2	0	0	0	0	9	0	0	0	0	0	0	0	0
0	9	2	0	0	0	0	9	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	9	2	0	0	0	0	9
0	0	0	0	0	0	0	0	0	9	2	0	0	0	0	9
2	0	0	0	0	9	0	9	0	3	0	0	0	0	8	8
2	0	0	0	0	9	0	9	0	3	0	0	0	0	8	8
0	0	0	8	8	0	3	2	0	0	0	0	0	6	1	
0	0	0	8	8	0	3	2	0	0	0	0	0	6	1	
0	0	0	0	6	1	2	0	0	0	0	0	0	0	0	
0	0	0	0	6	1	2	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	6	1	2	0	
0	0	0	0	0	0	0	0	0	0	0	6	1	2	0	
0	0	6	1	2	0	0	0	7	0	3	0	0	b	0	0
0	0	6	1	2	0	0	0	7	0	3	0	0	b	0	0
3	0	0	b	0	0	7	0	0	2	0	1	7	0	0	0
3	0	0	b	0	0	7	0	0	2	0	1	7	0	0	0
0	1	7	0	0	0	2	0	0	0	0	0	0	0	0	0
0	1	7	0	0	0	2	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	7	0	0	0	0	2	
0	0	0	0	0	0	0	0	1	7	0	0	0	0	2	
7	0	0	0	2	0	1	0	b	0	0	0	0	a	d	
7	0	0	0	2	0	1	0	b	0	0	0	0	a	d	
0	0	0	0	a	4	0	b	7	0	0	0	0	8	7	
0	0	0	0	a	4	0	b	7	0	0	0	0	8	7	
0	0	0	0	8	7	7	0	0	0	0	0	0	0	0	
0	0	0	0	8	7	7	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	8	7	7	0	
0	0	0	0	0	0	0	0	0	0	0	8	7	7	0	
0	0	8	7	7	0	0	1	0	b	0	0	c	0	0	
0	0	8	7	7	0	0	1	0	b	0	0	c	0	0	
b	0	0	c	0	0	1	0	0	8	0	7	0	0	0	
b	0	0	c	0	0	1	0	0	8	0	7	0	0	0	
0	7	0	0	0	0	8	b	0	0	0	0	0	0	0	
0	7	0	0	0	0	8	b	0	0	0	0	0	0	0	
0	0	0	0	0	0	b	0	0	7	0	0	c	0	0	8
0	0	0	0	0	0	b	0	0	7	0	0	c	0	0	8
0	7	0	0	c	0	0	8	0	0	0	0	0	0	b	0
0	7	0	0	c	0	0	8	0	0	0	0	0	0	b	0

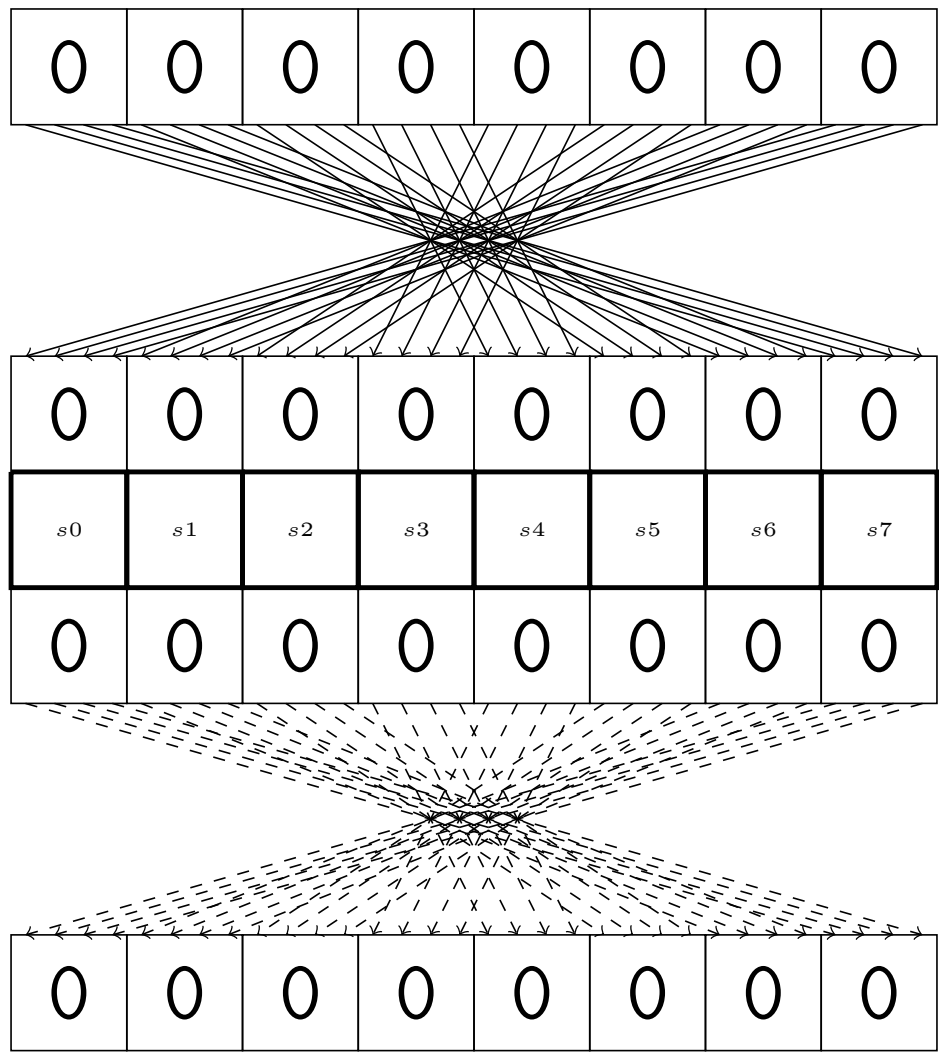
2 LBlock_round

Maximal linear correlation: $2^{-0.0}$



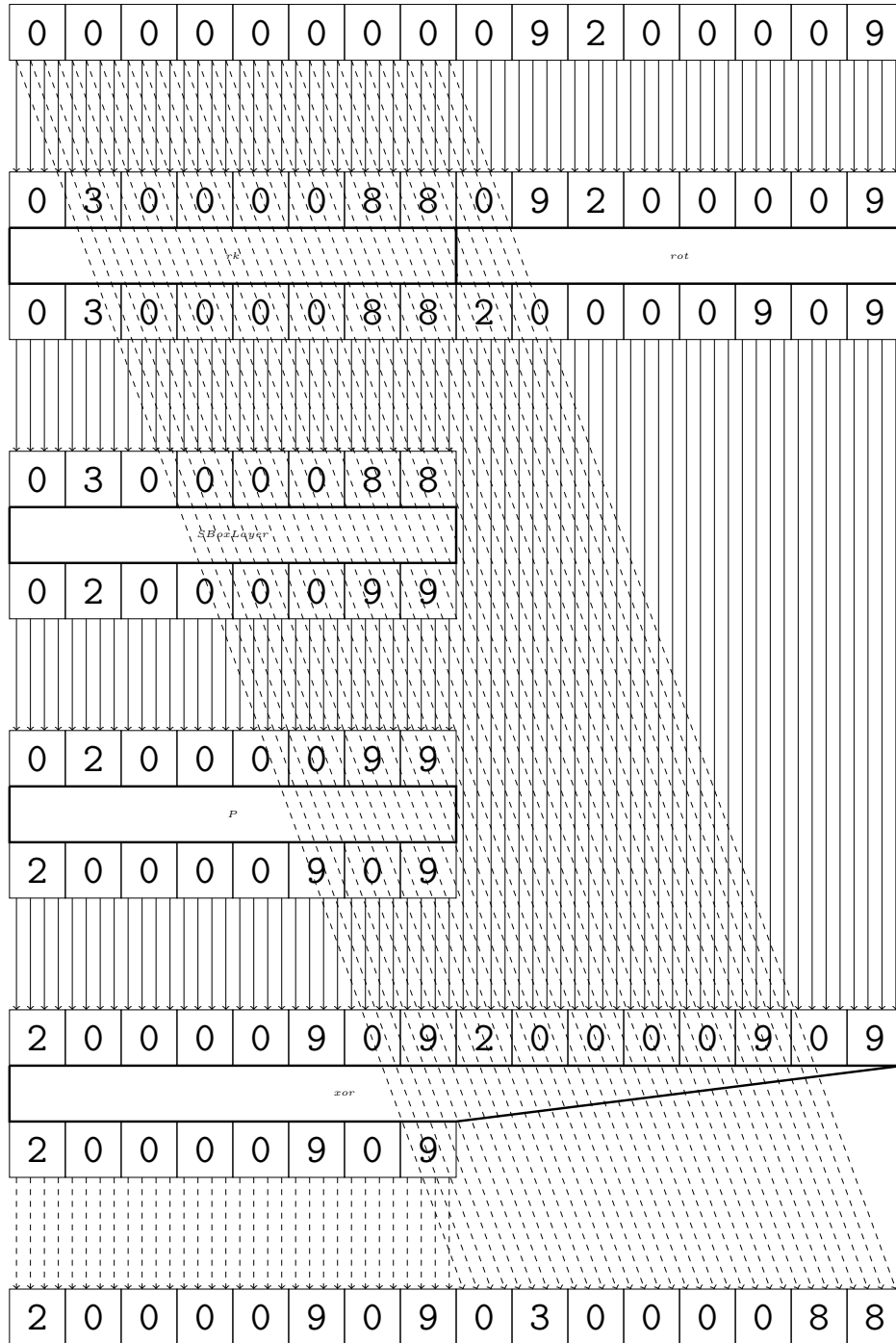
3 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



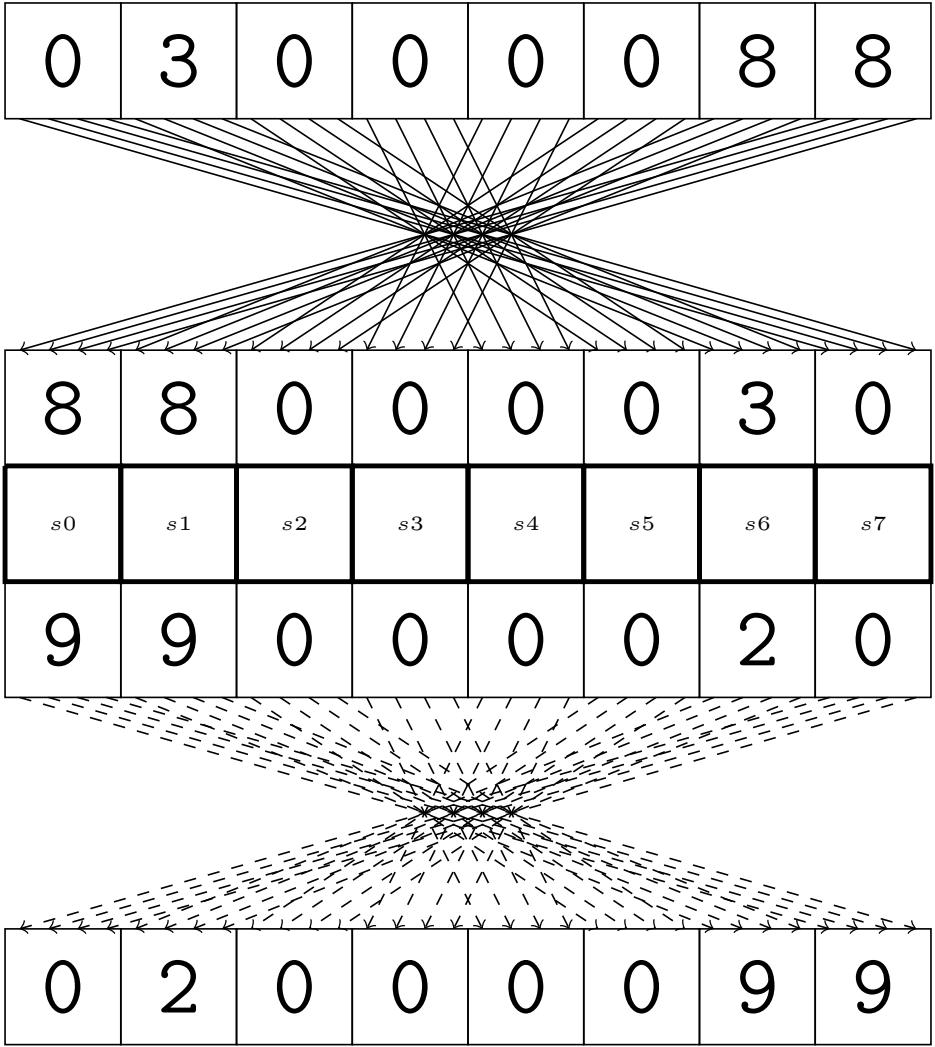
4 LBlock_round

Maximal linear correlation: $2^{-3.0}$



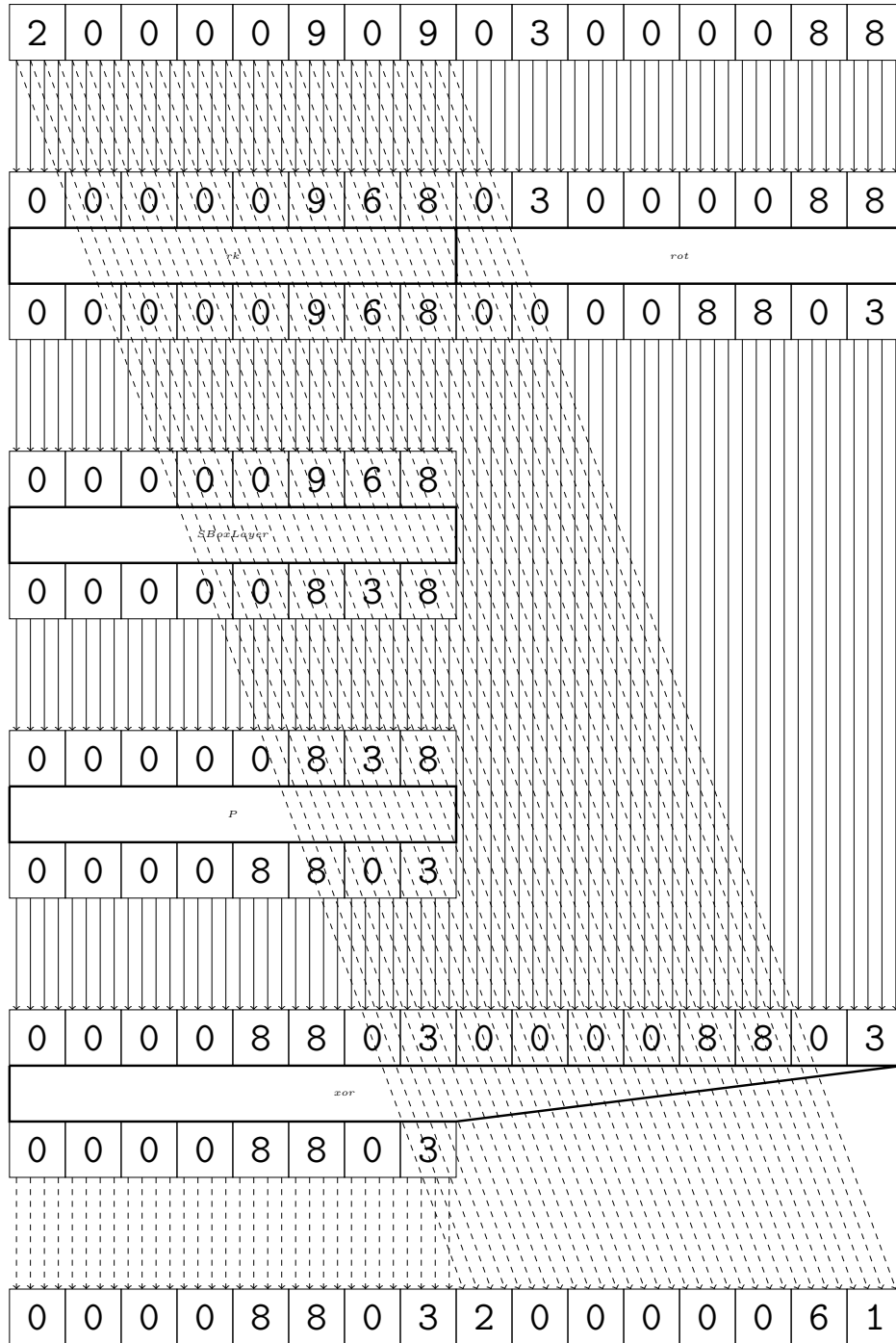
5 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



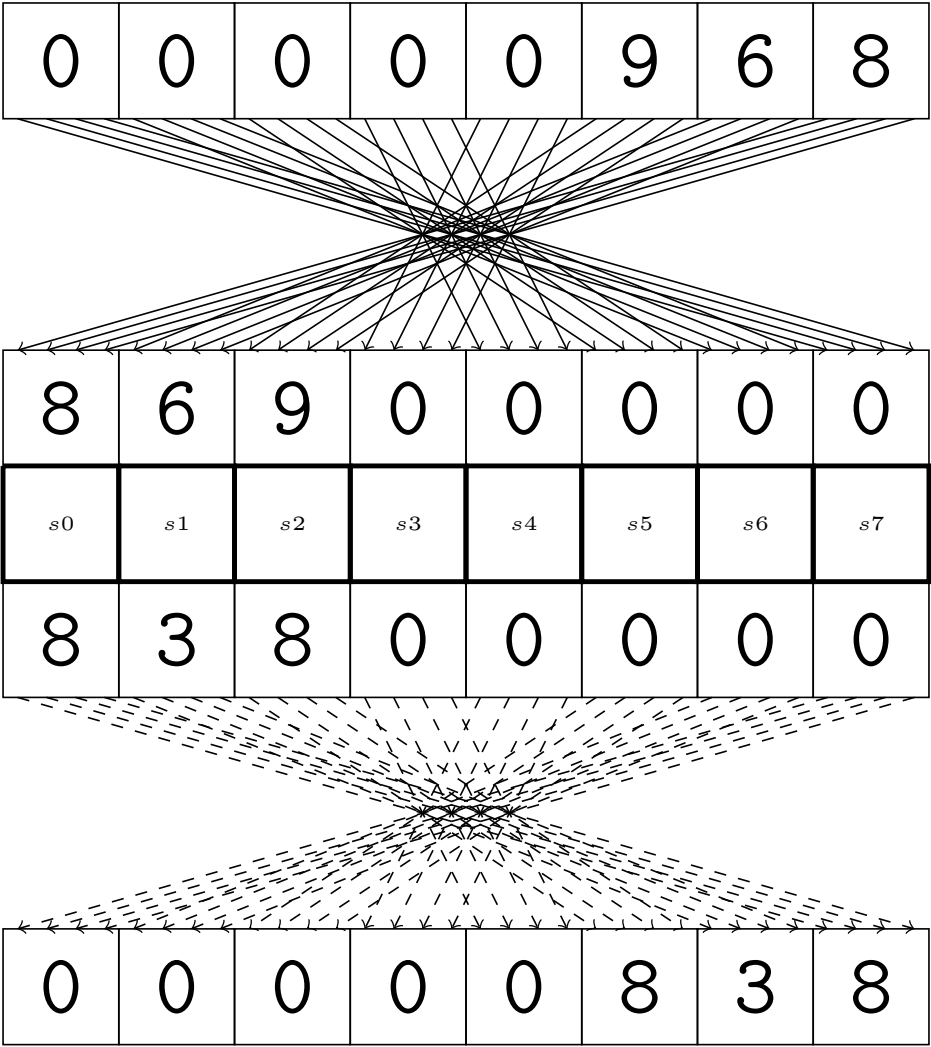
6 LBlock_round

Maximal linear correlation: $2^{-3.0}$



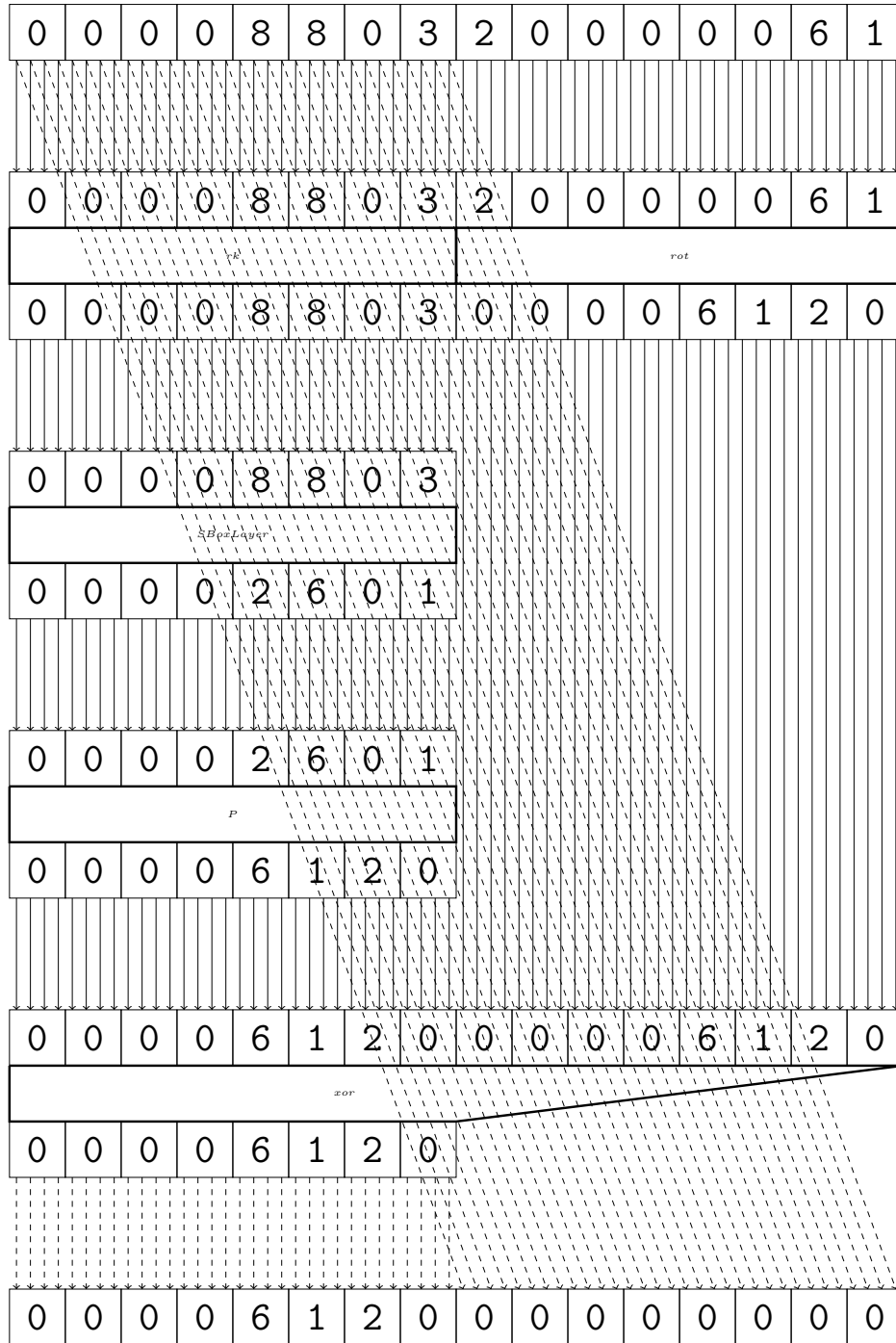
7 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



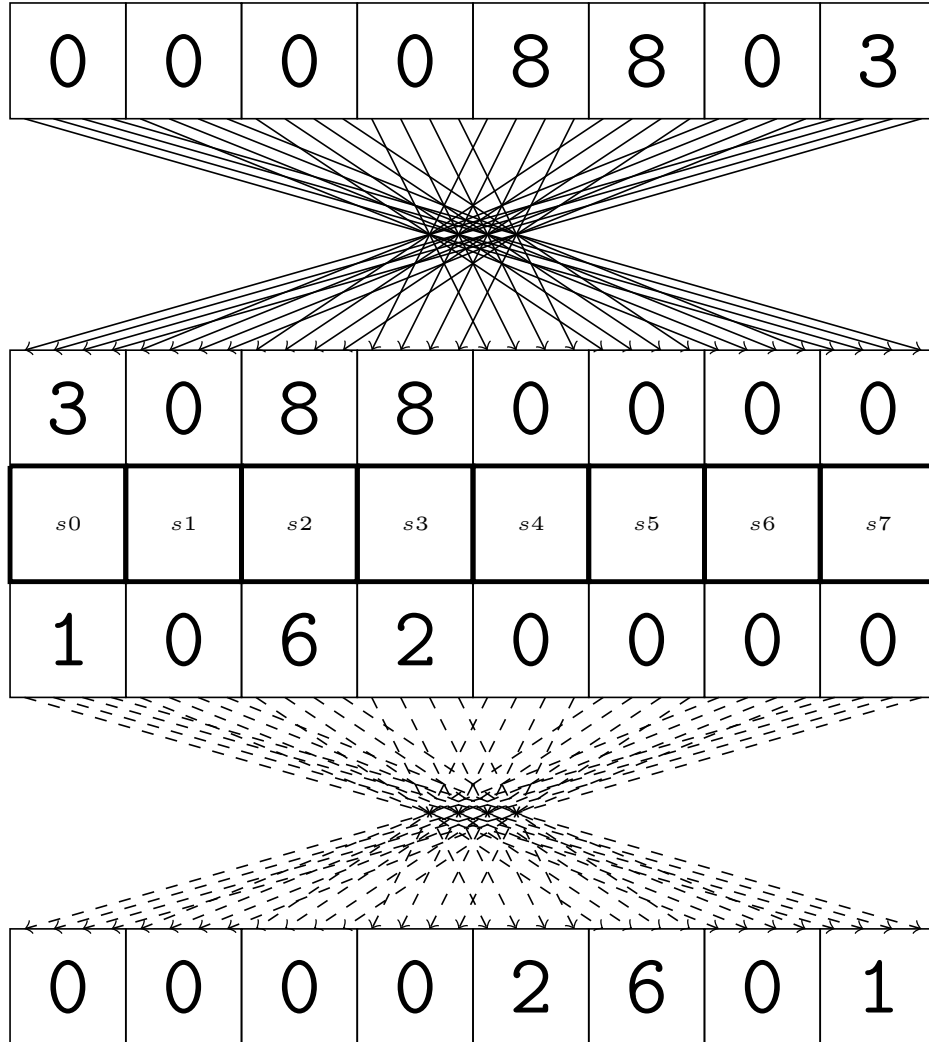
8 LBlock_round

Maximal linear correlation: $2^{-3.0}$

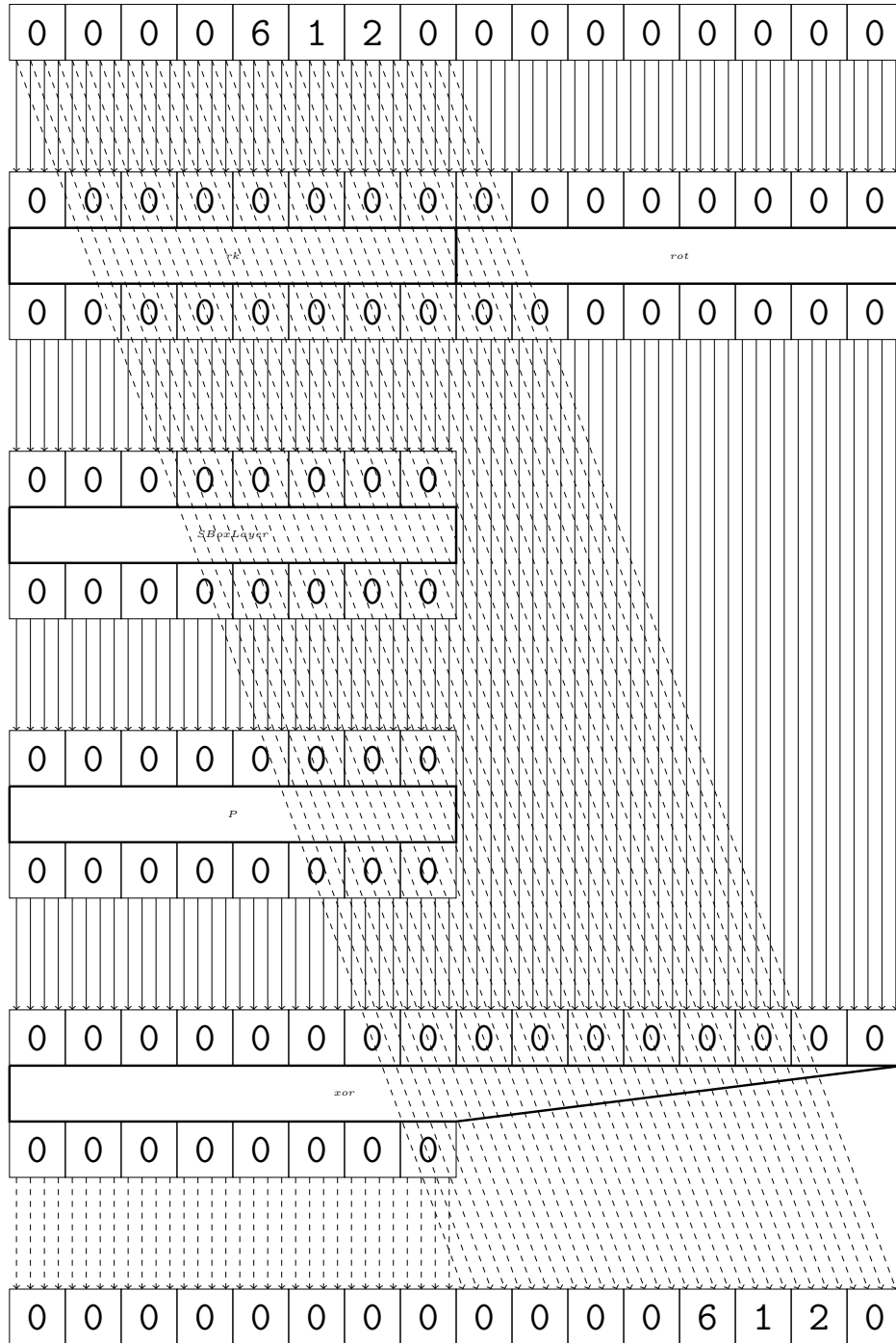


9 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

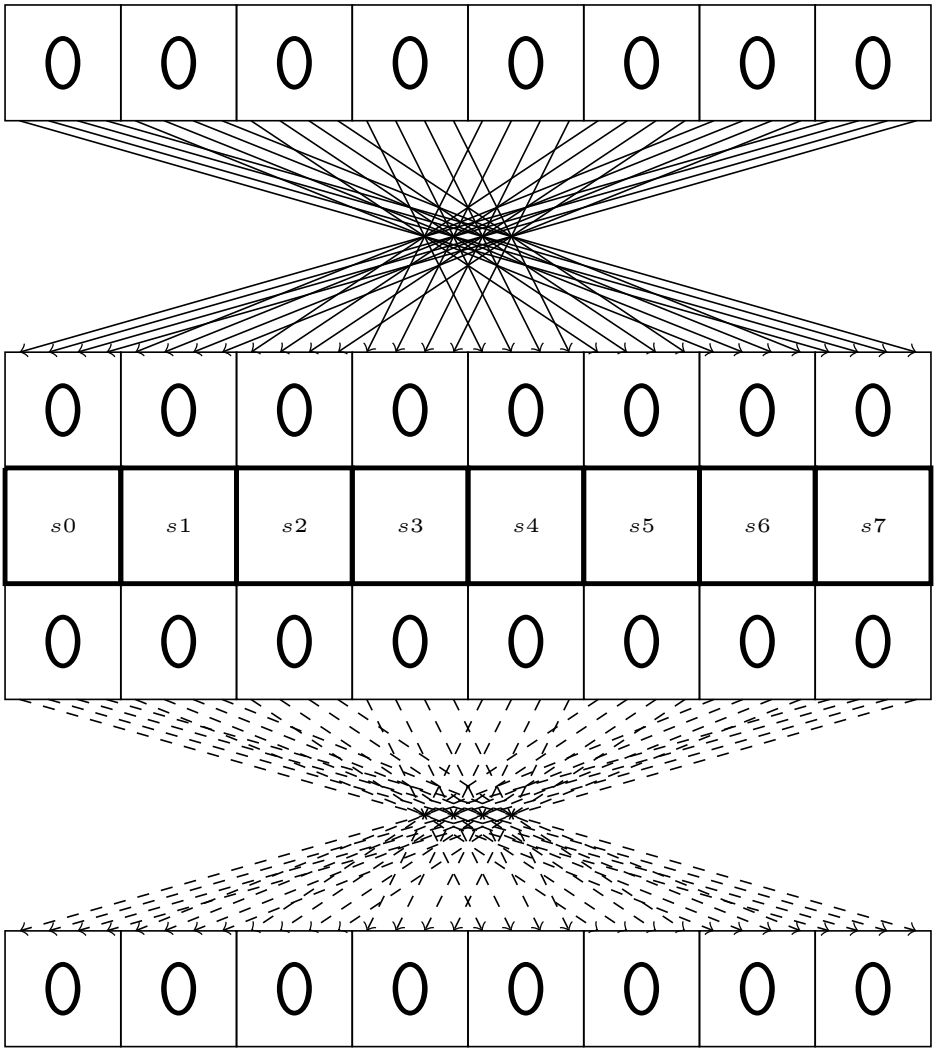


10 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

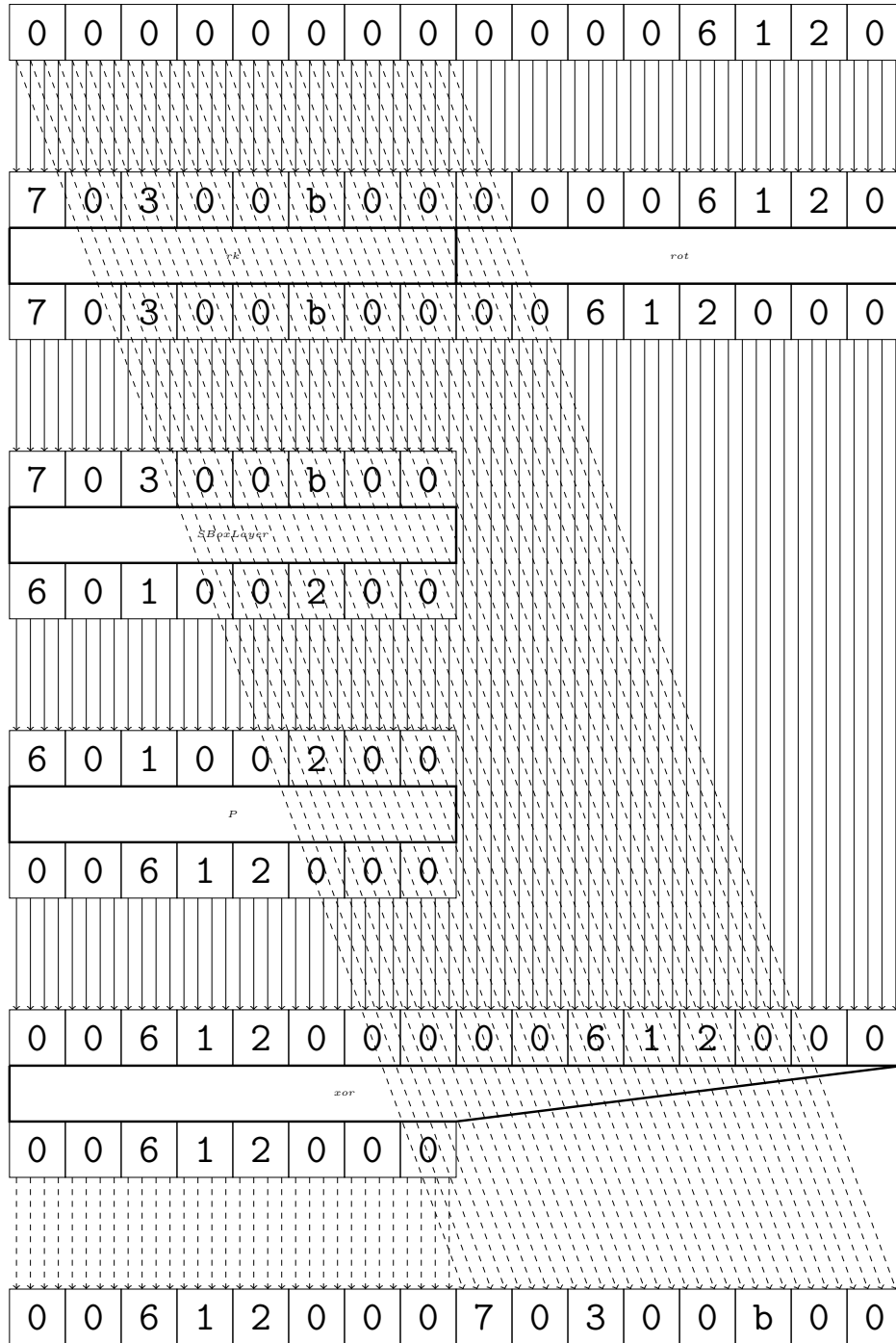
11 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



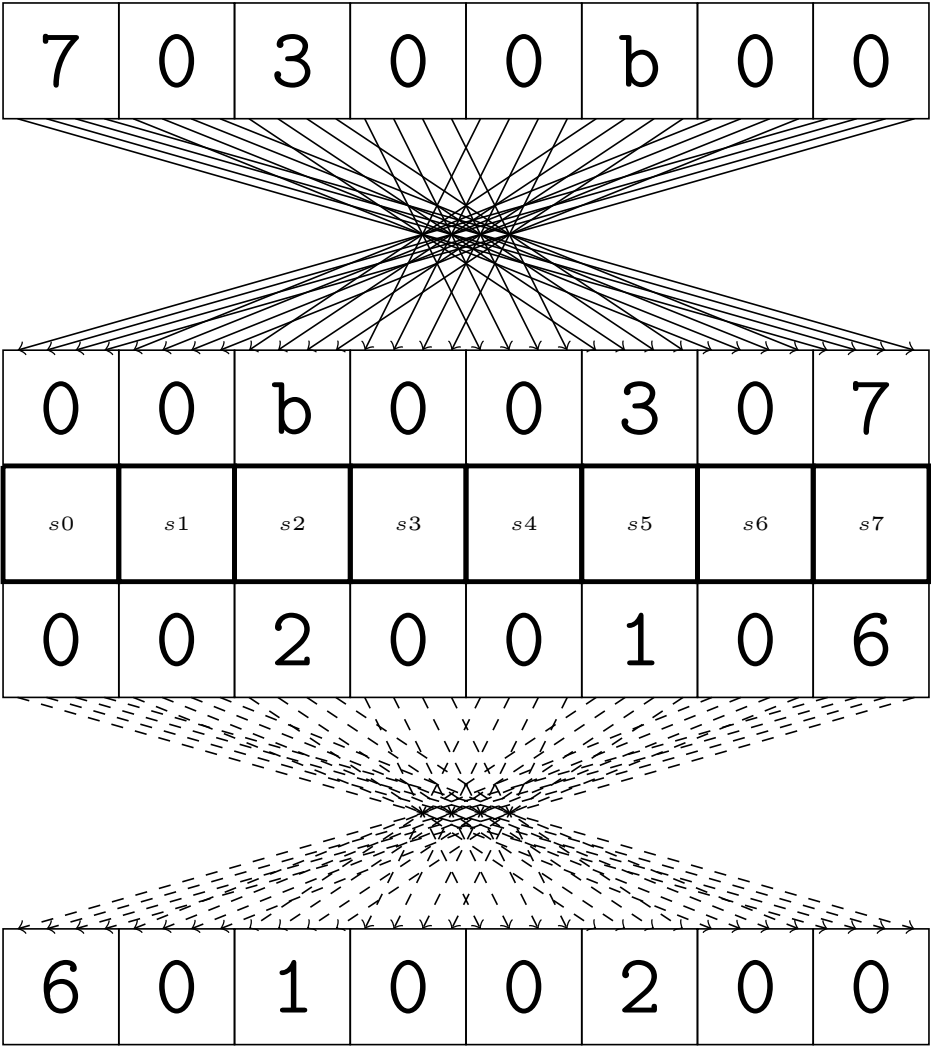
12 LBlock_round

Maximal linear correlation: $2^{-3.0}$

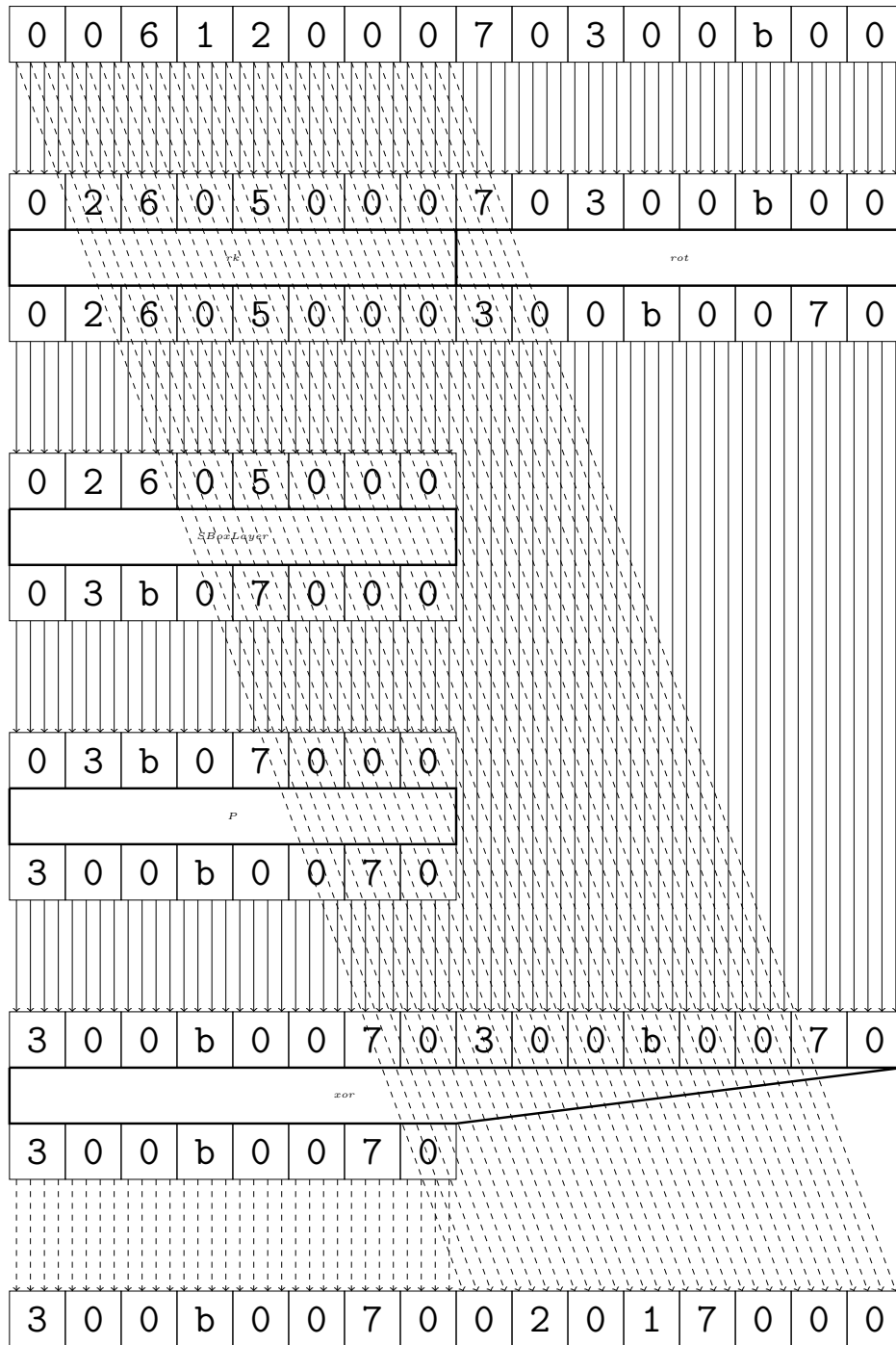


13 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

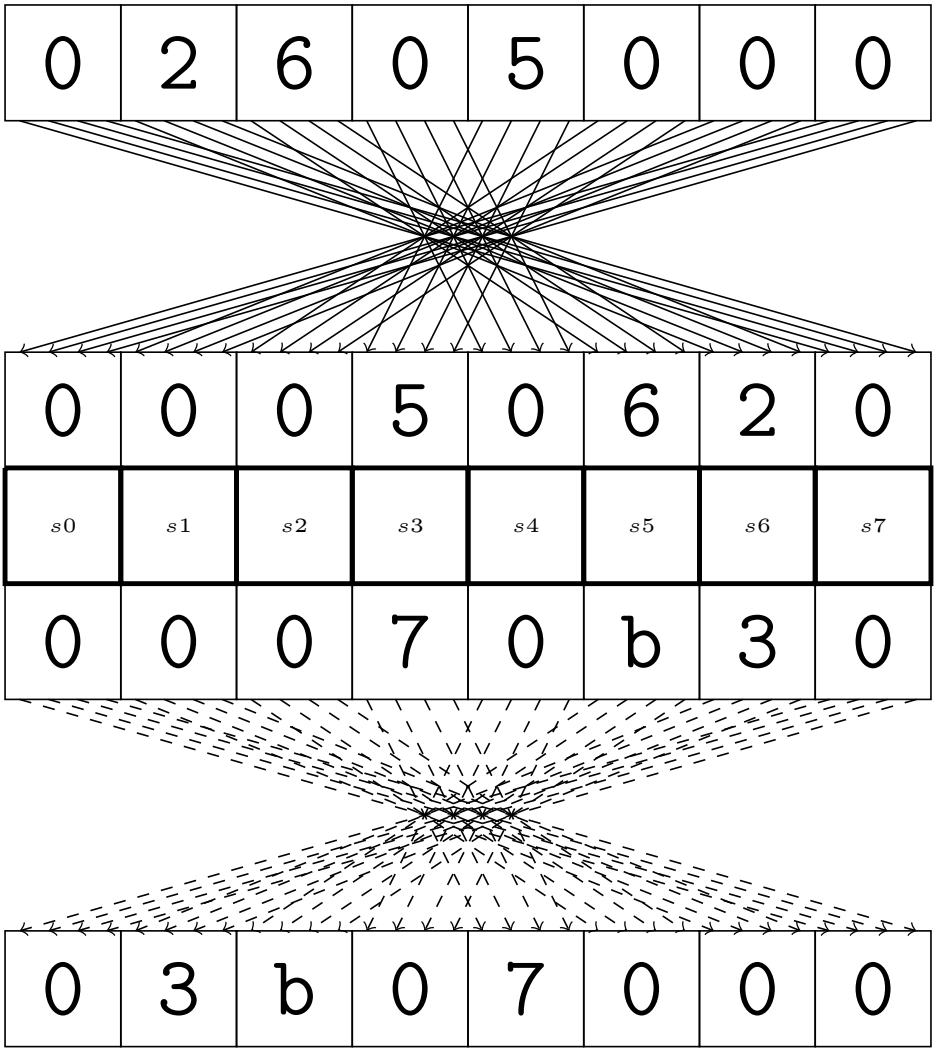


14 LBlock_round

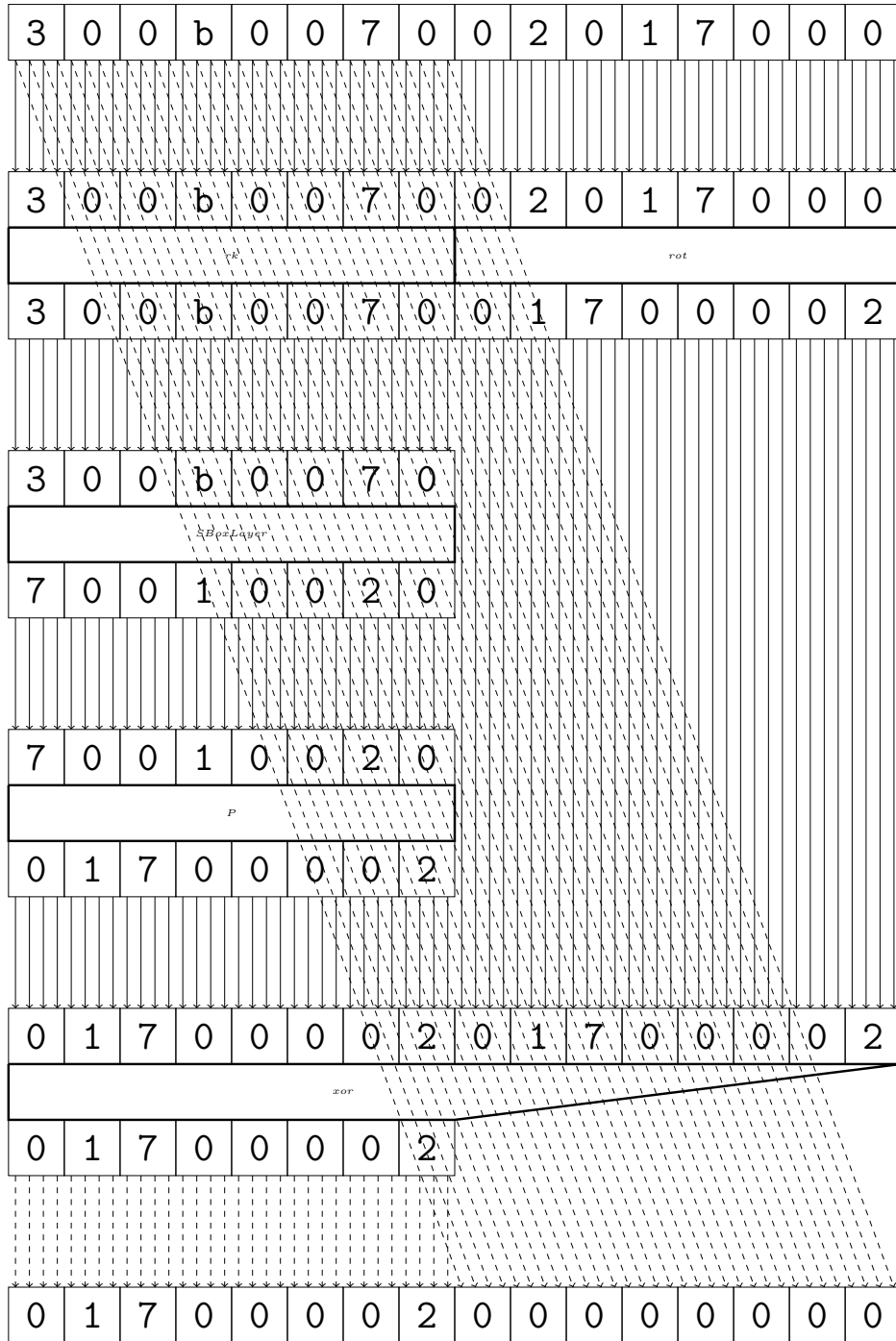
Maximal linear correlation: $2^{-3.0}$ 

15 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

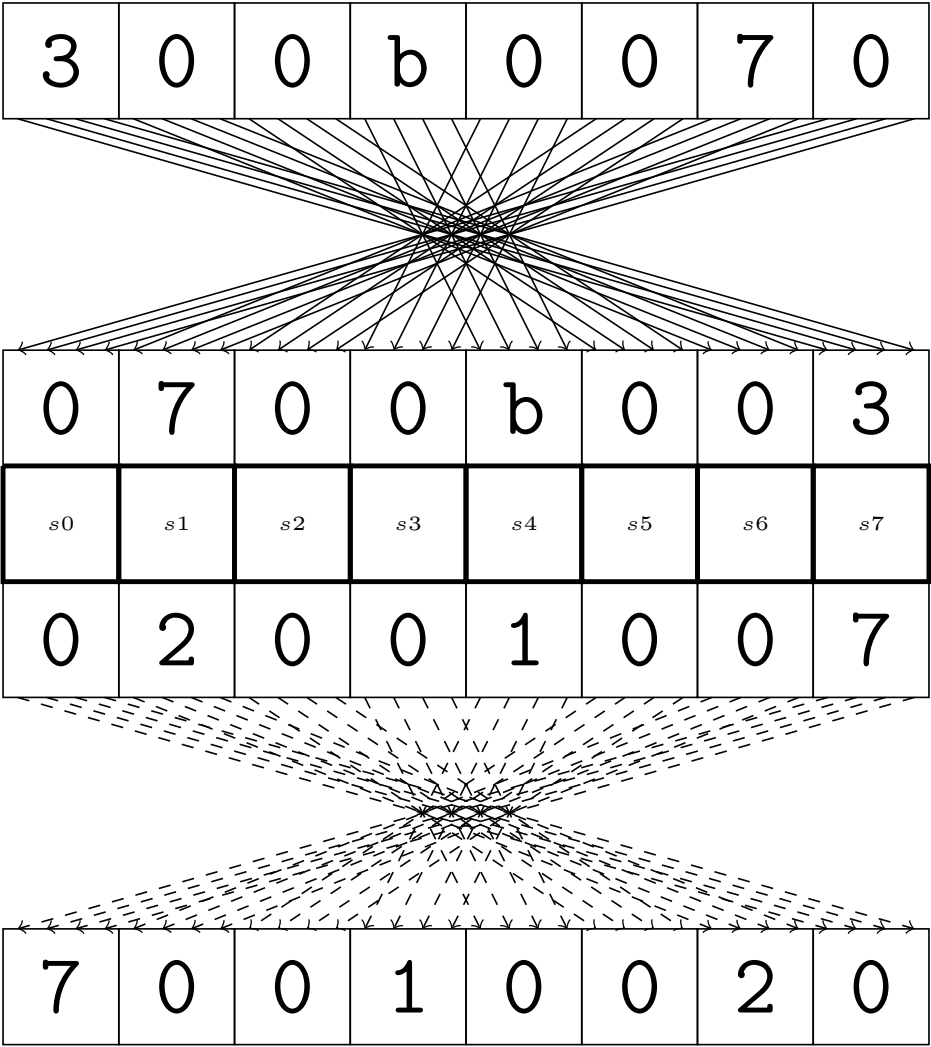


16 LBlock_round

Maximal linear correlation: $2^{-3.0}$ 

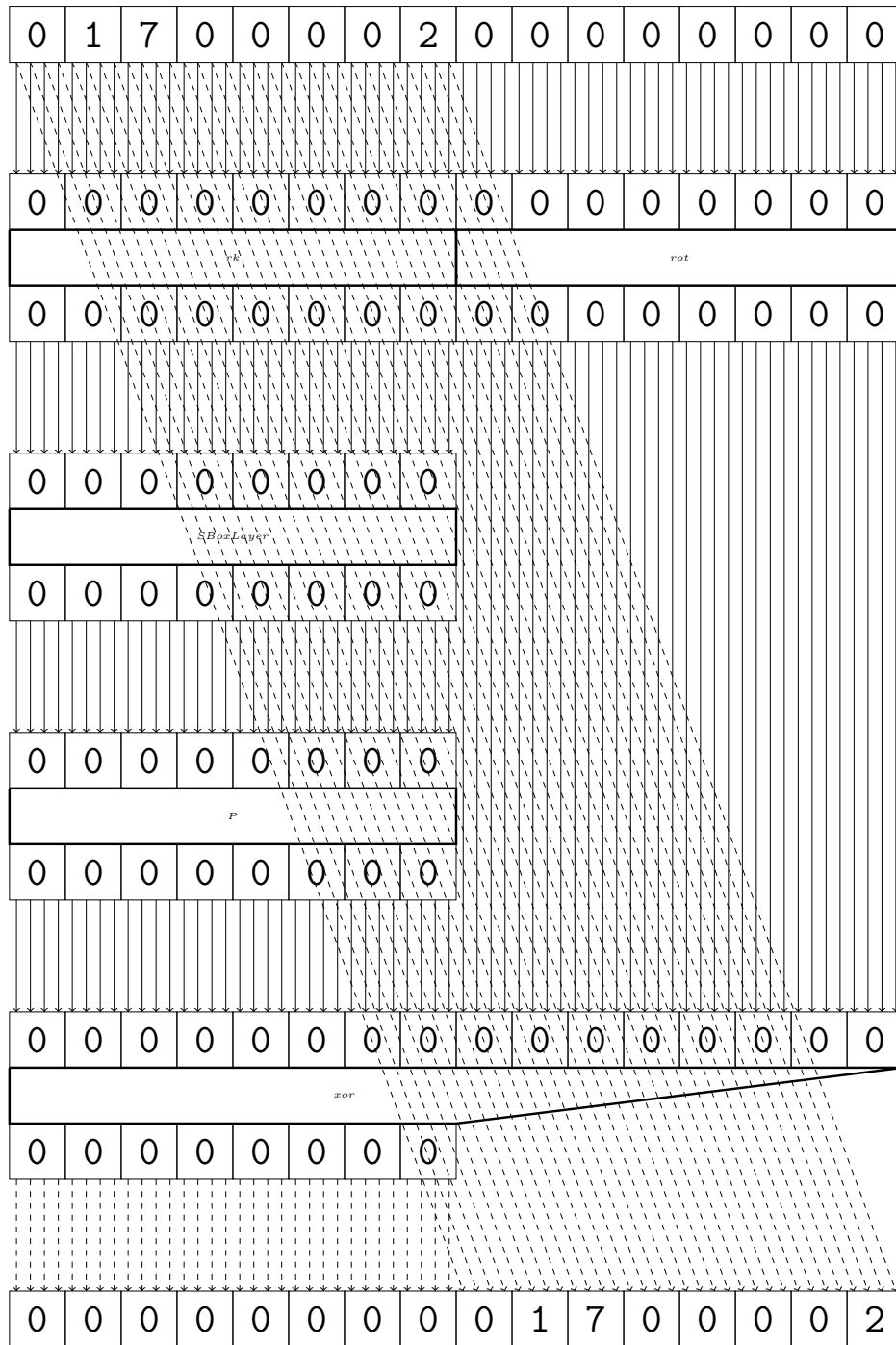
17 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



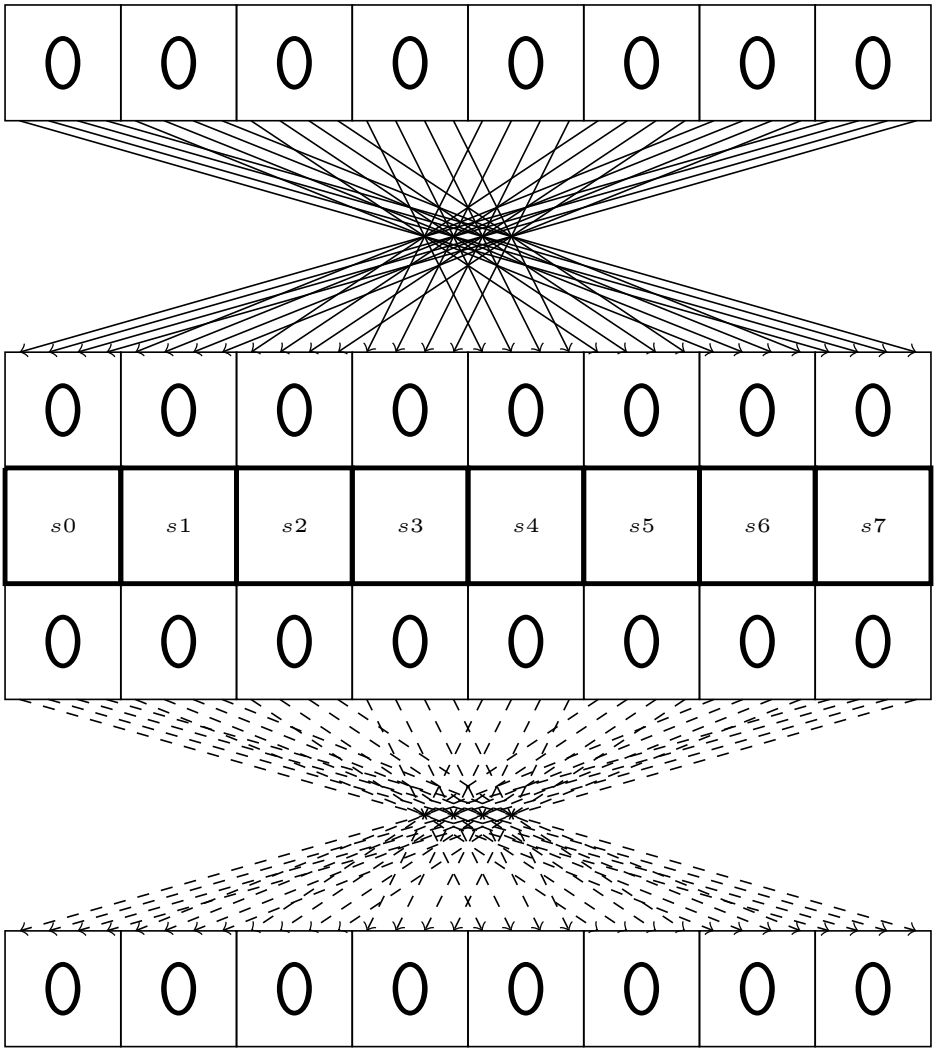
18 LBlock_round

Maximal linear correlation: $2^{-0.0}$



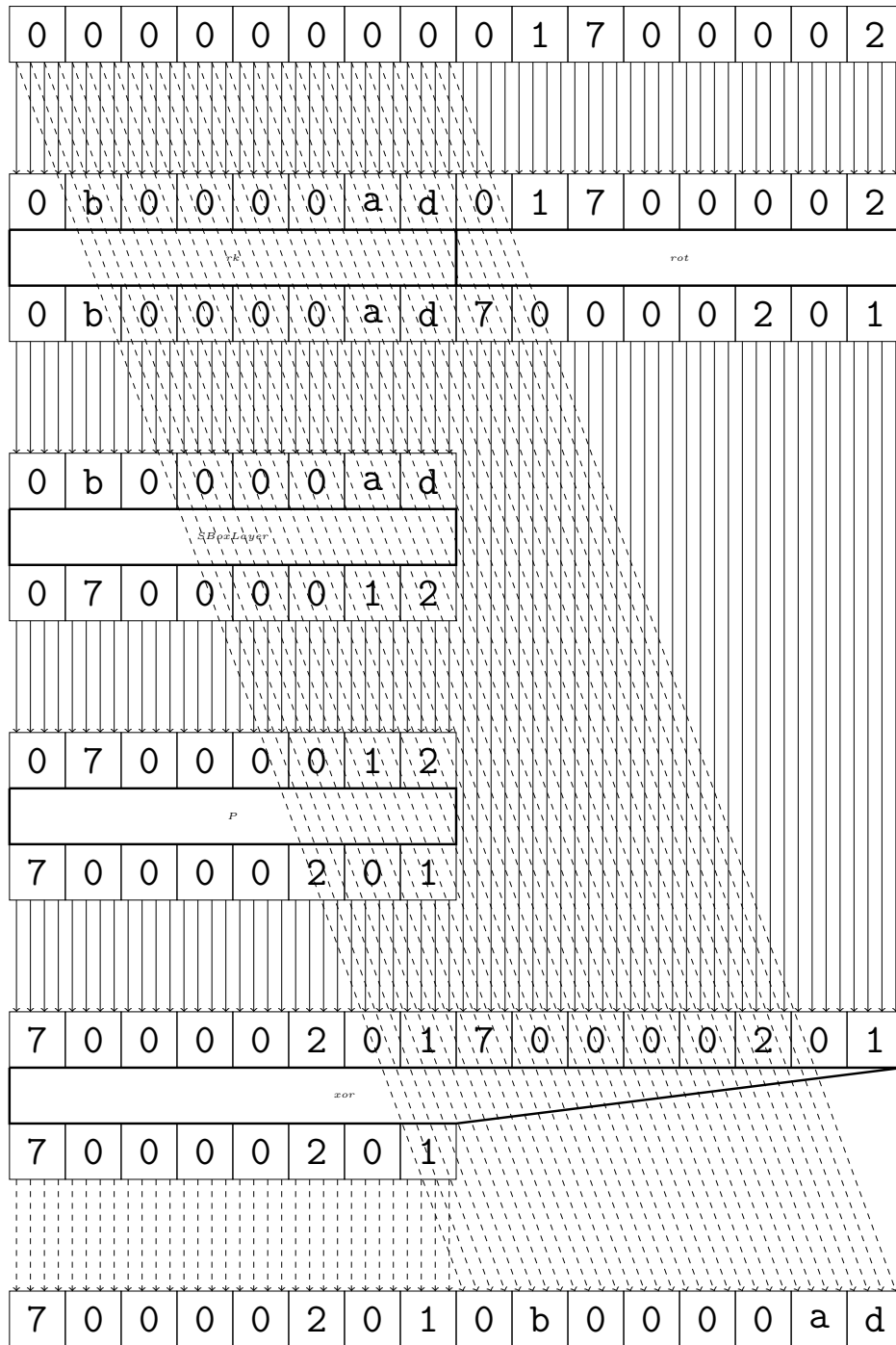
19 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



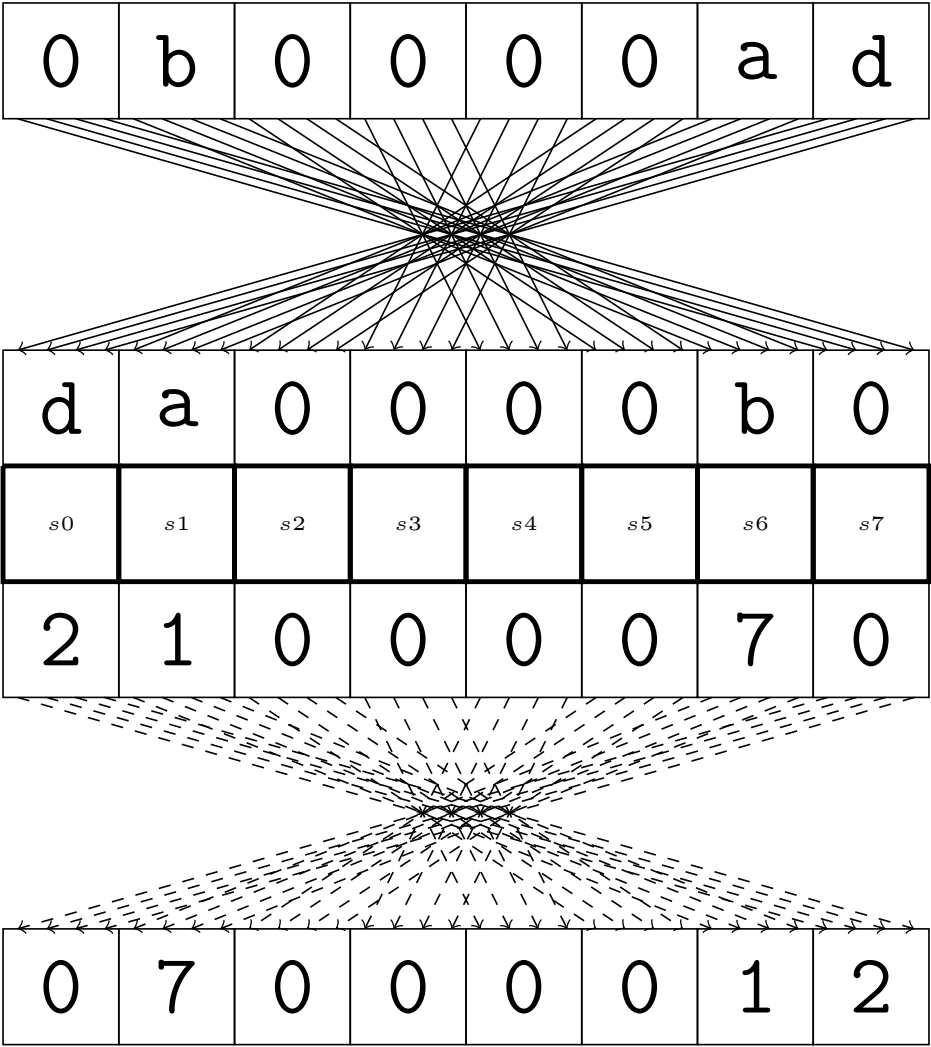
20 LBlock_round

Maximal linear correlation: $2^{-3.0}$



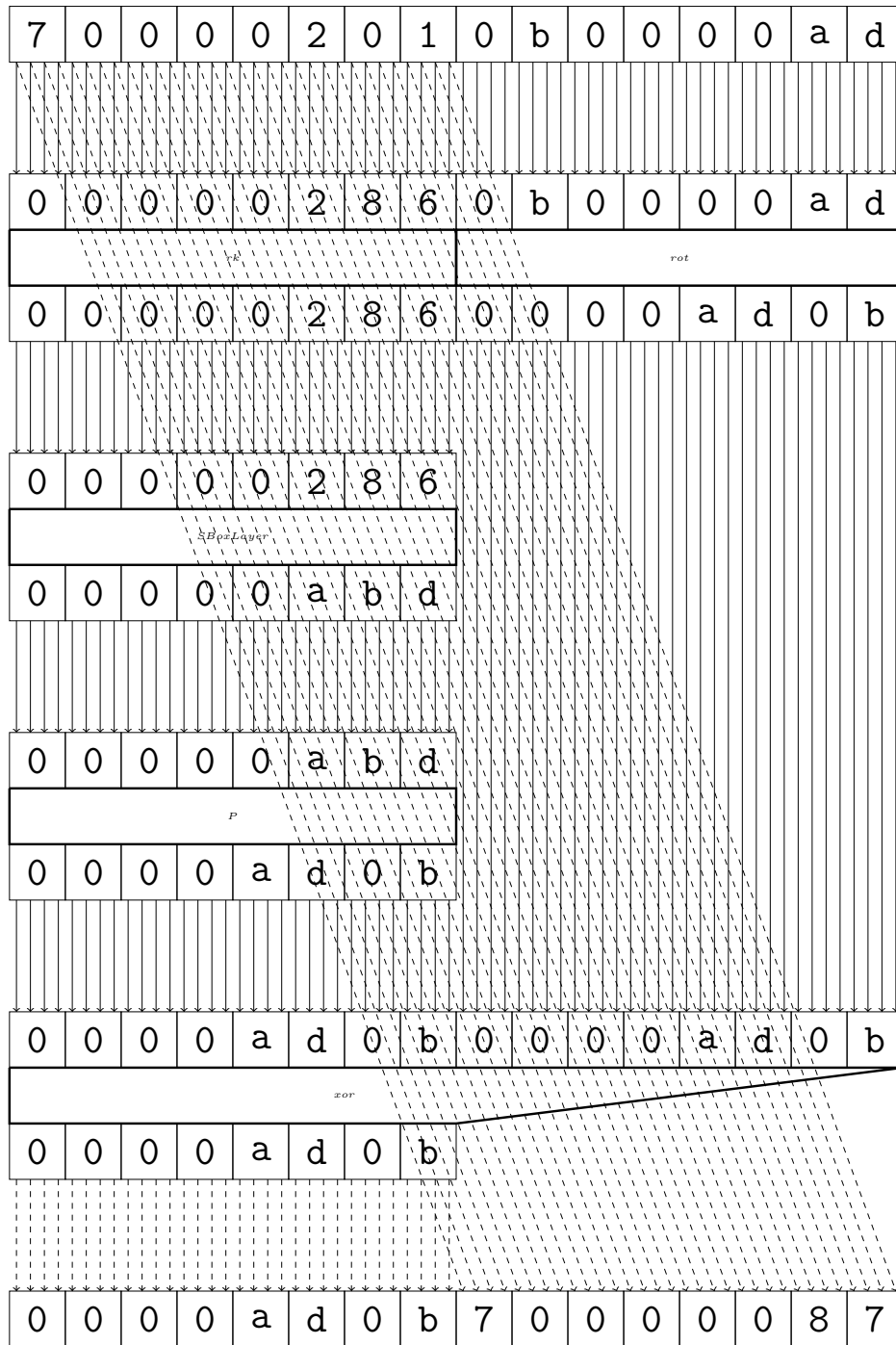
21 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



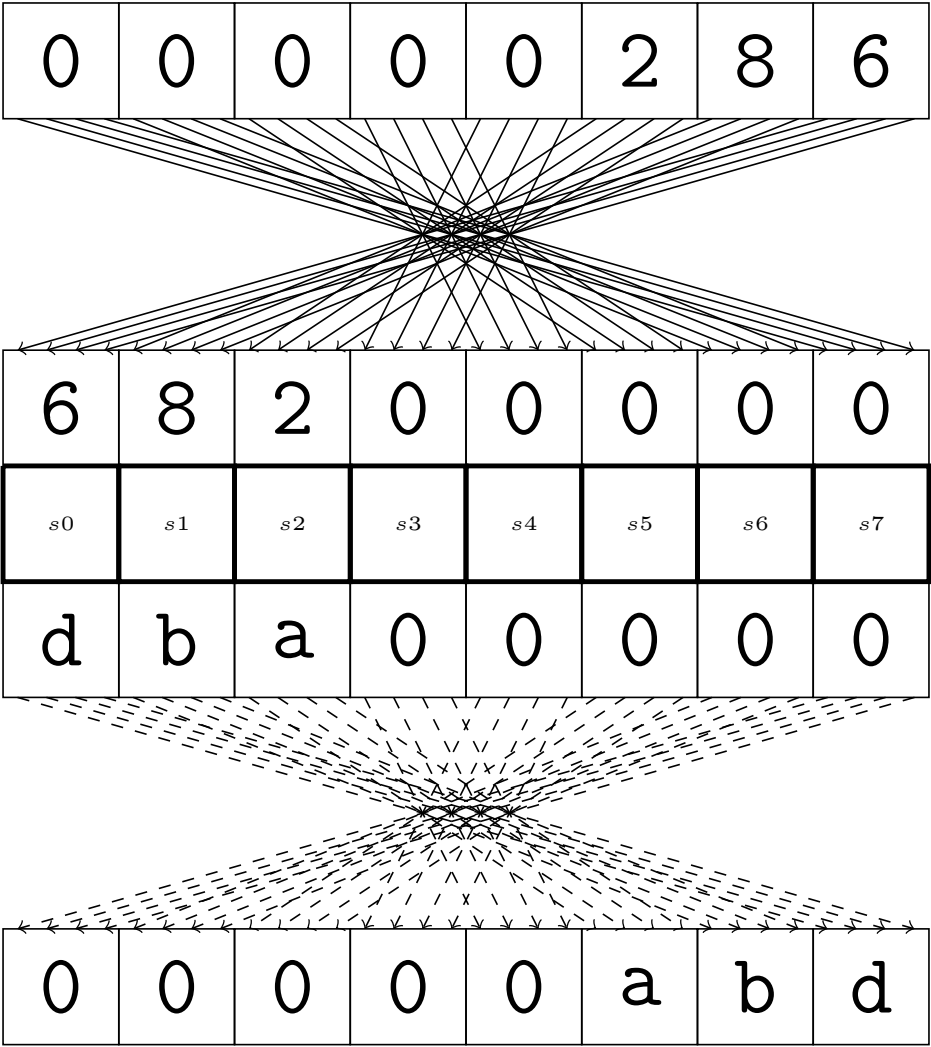
22 LBlock_round

Maximal linear correlation: $2^{-3.0}$

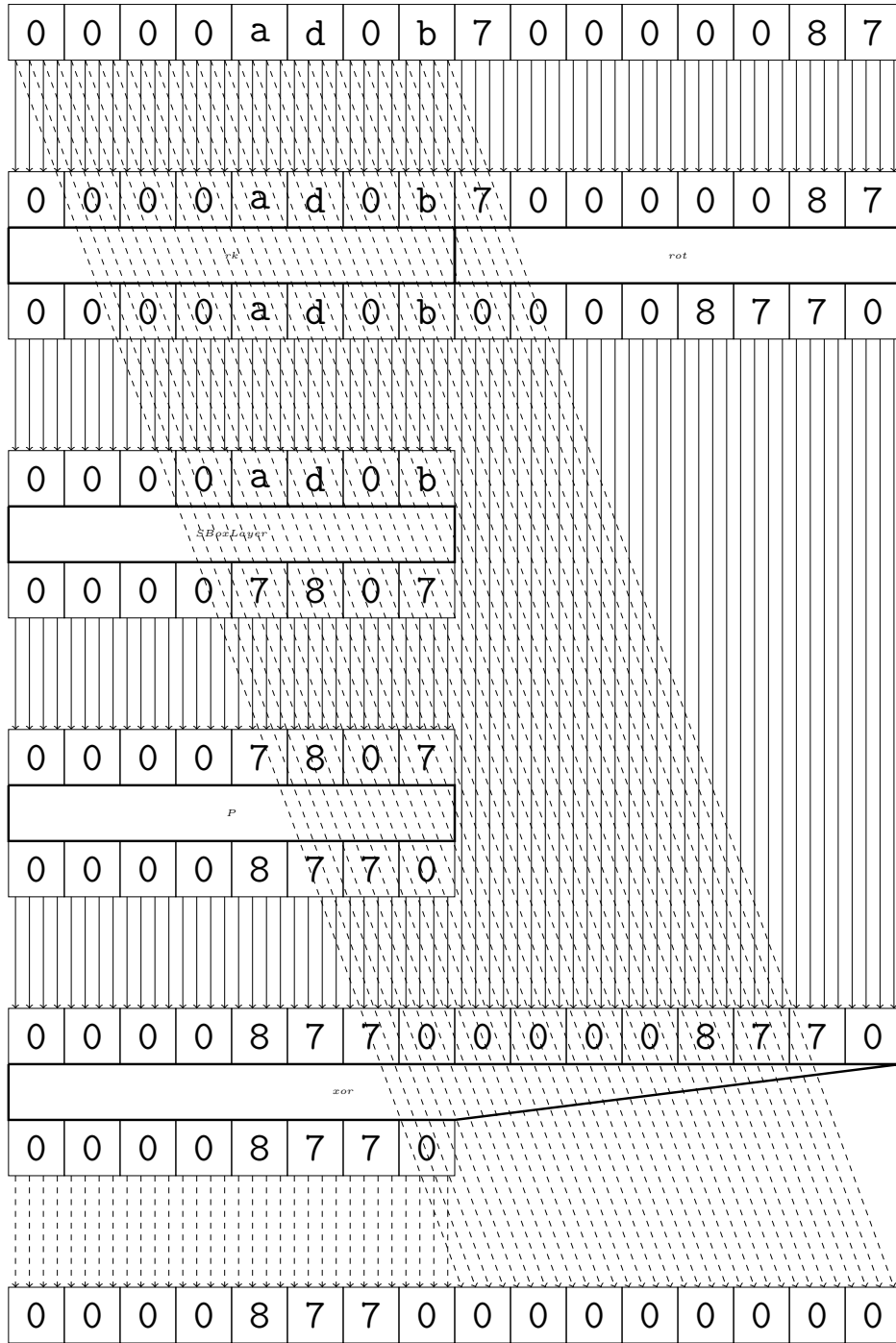


23 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

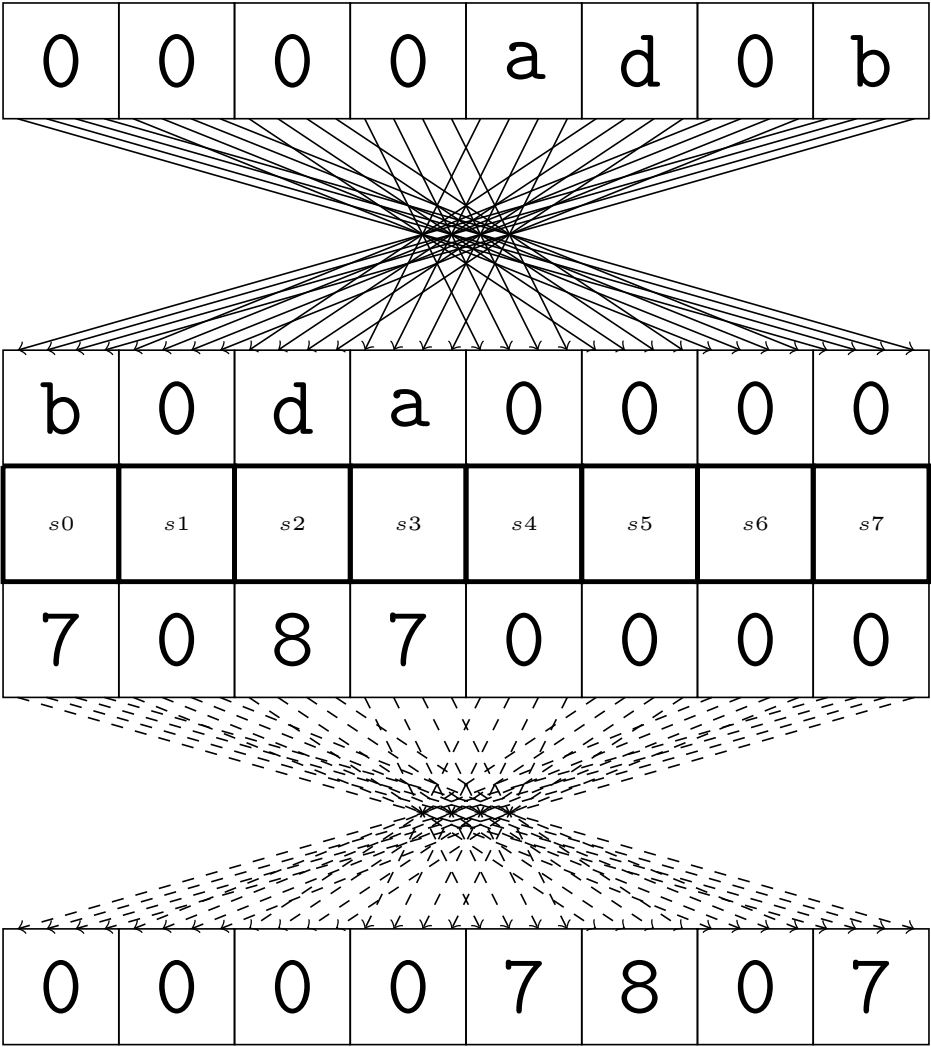


24 LBlock_round

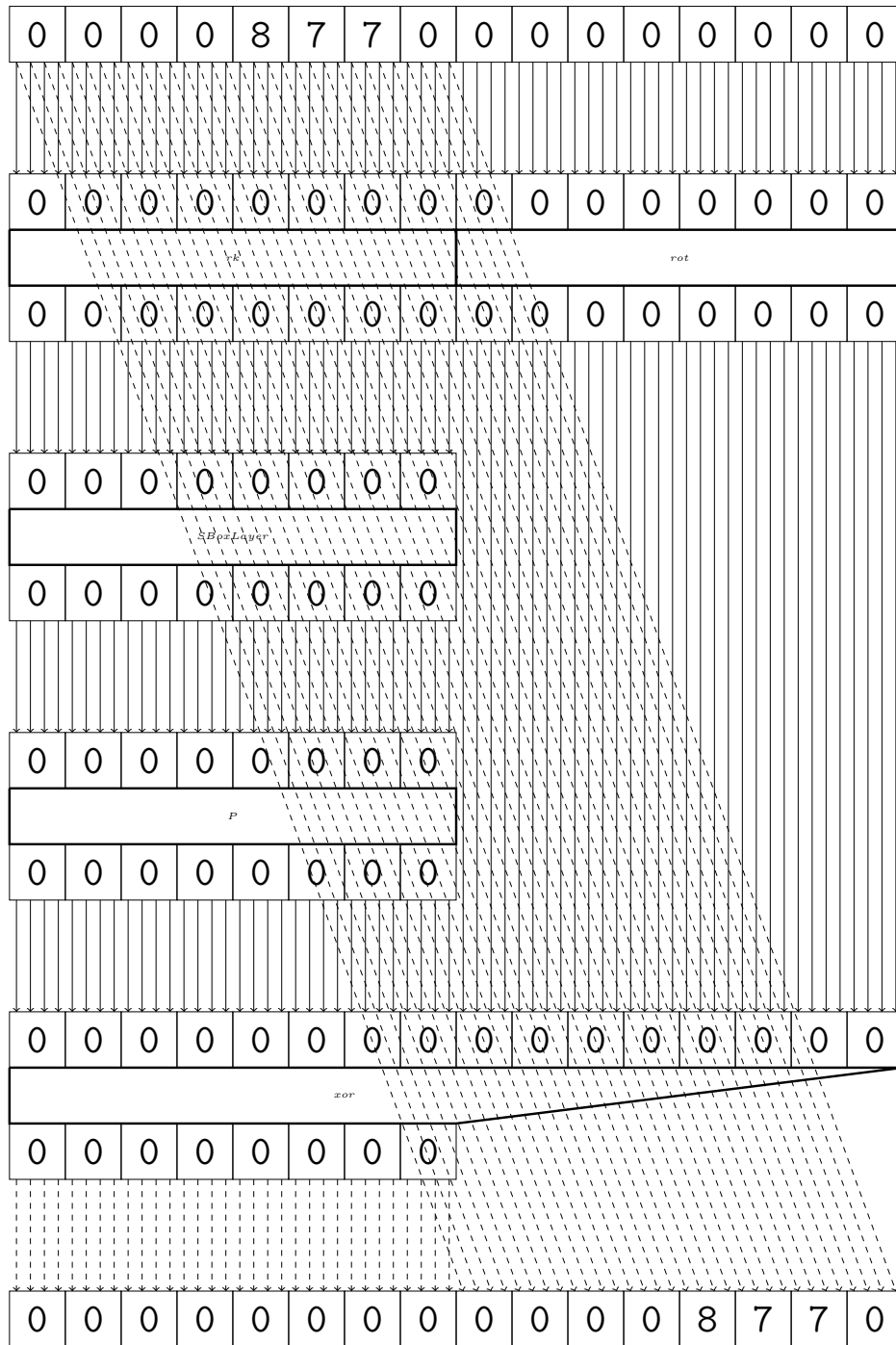
Maximal linear correlation: $2^{-4.0}$ 

25 SBoxLayer

Maximal linear correlation: $2^{-4.0}$

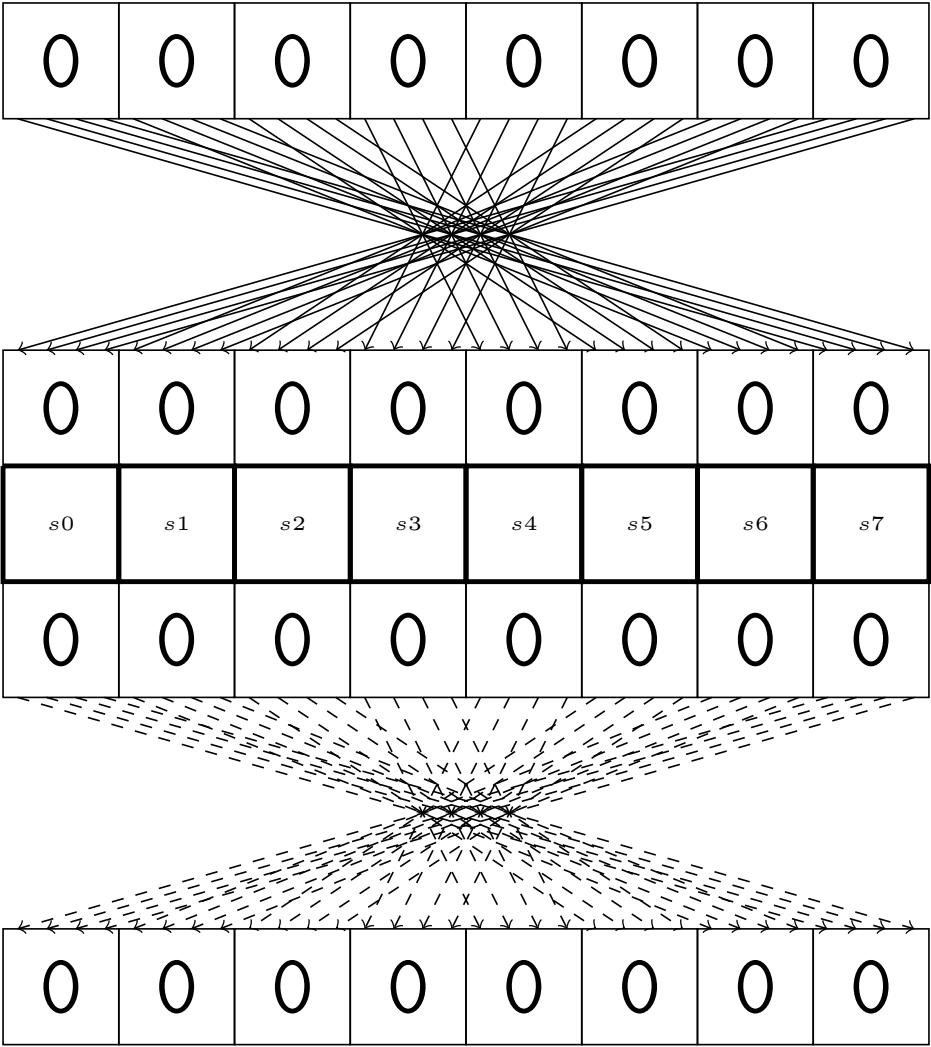


26 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

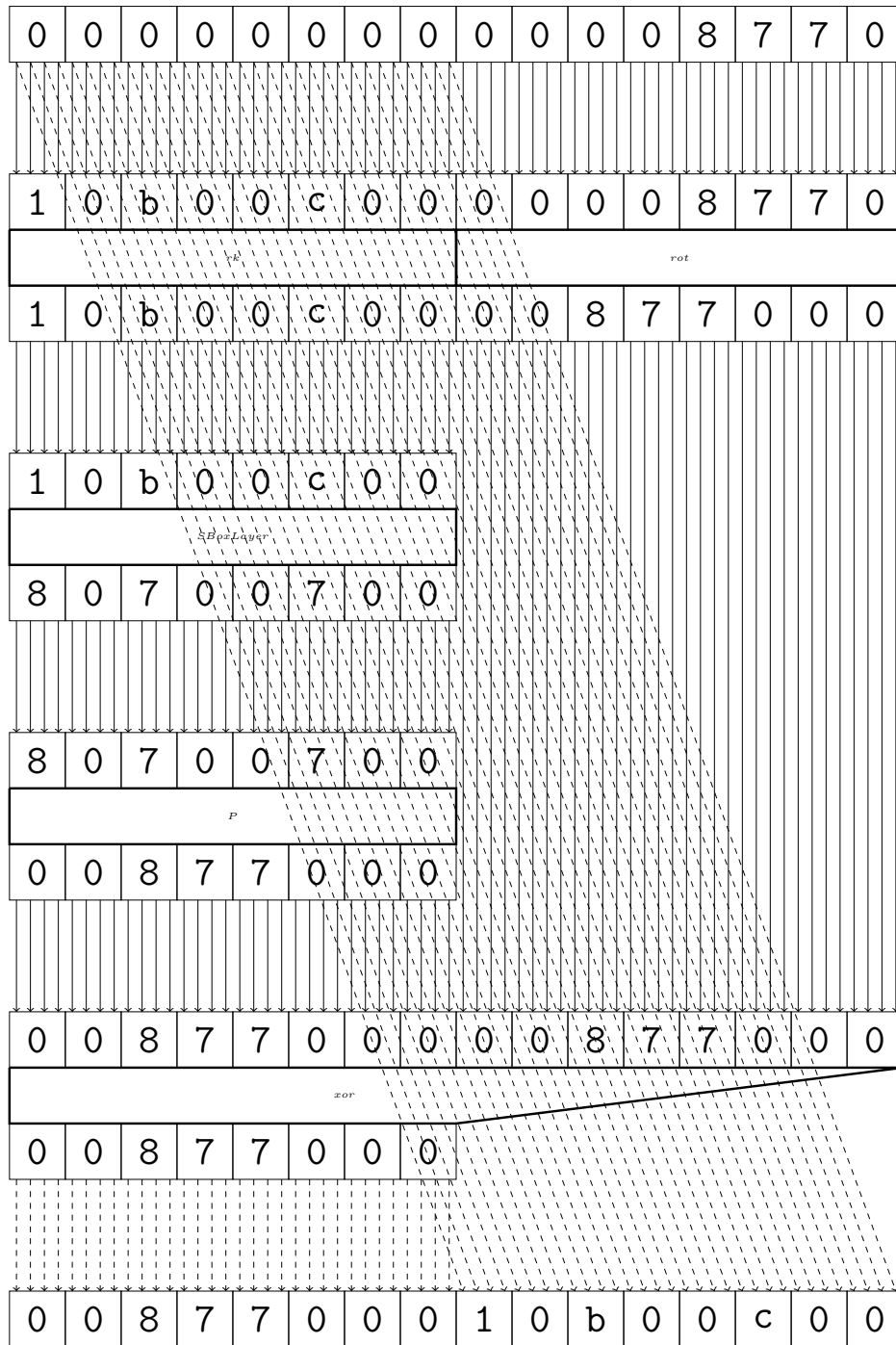
27 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



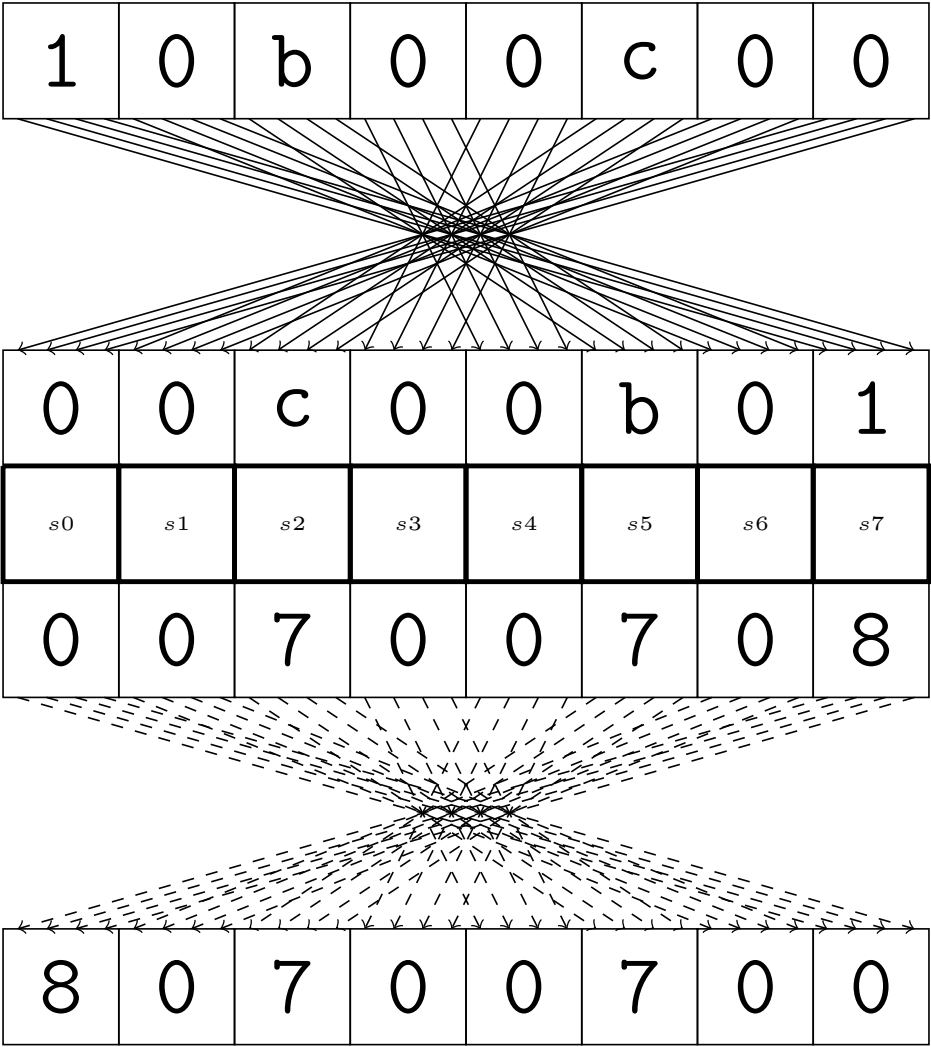
28 LBlock_round

Maximal linear correlation: $2^{-3.0}$



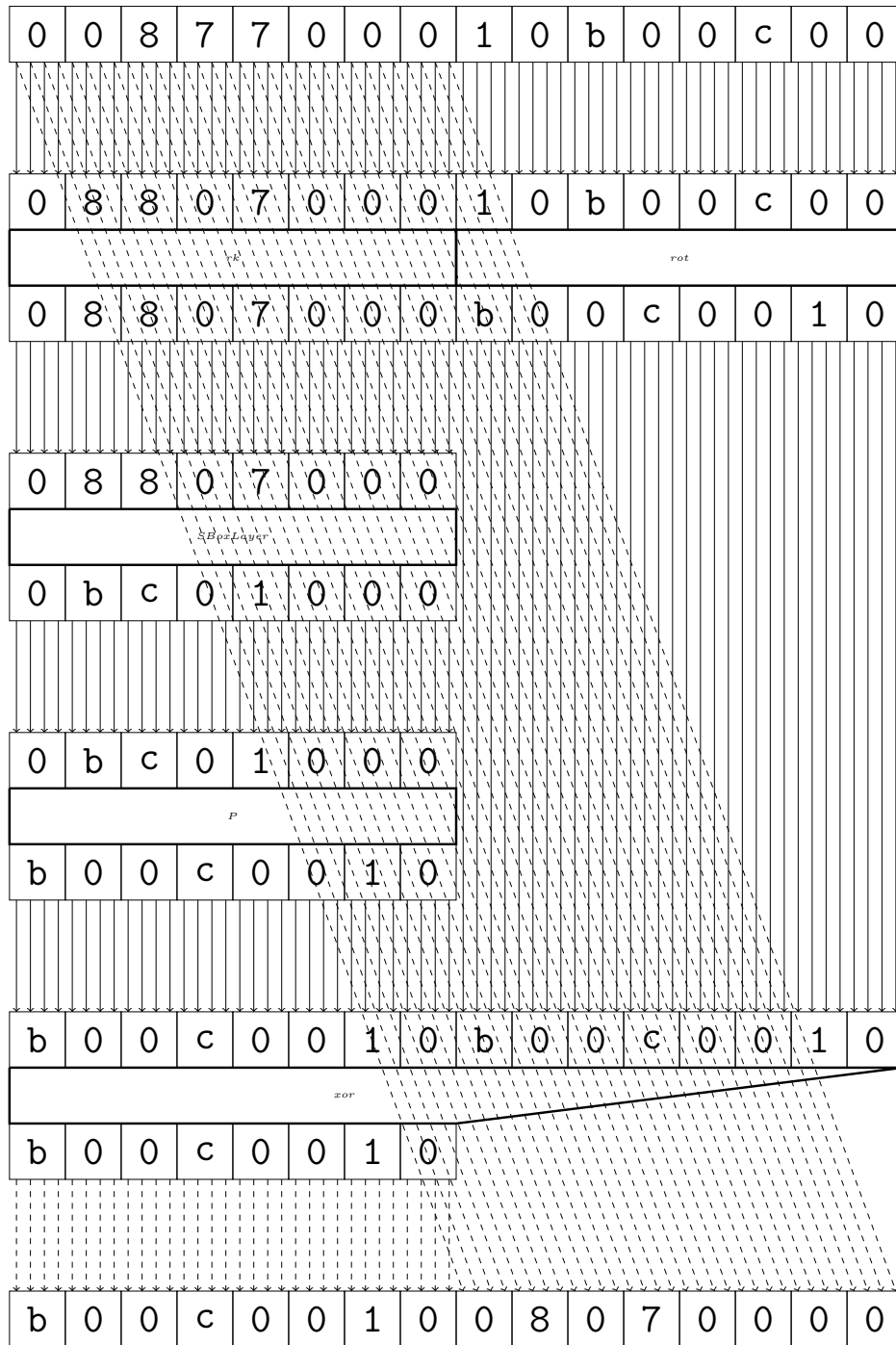
29 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



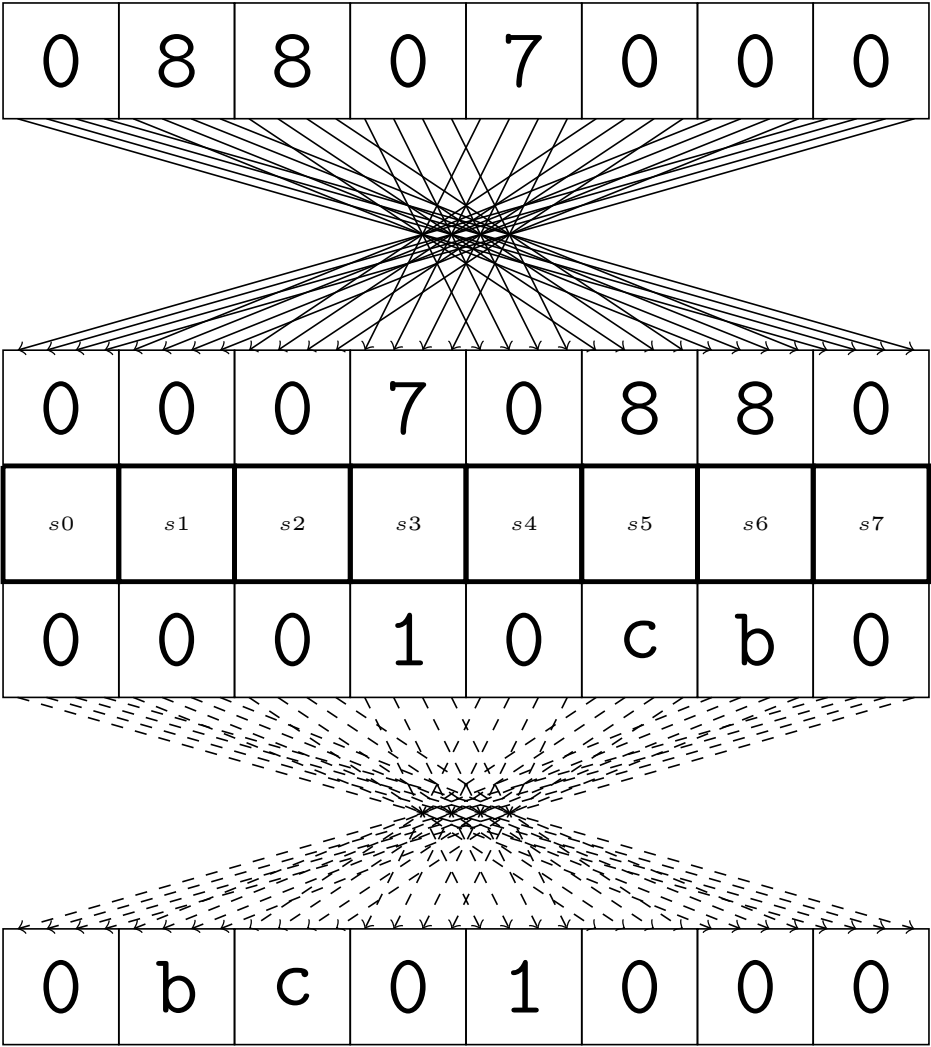
30 LBlock_round

Maximal linear correlation: $2^{-3.0}$

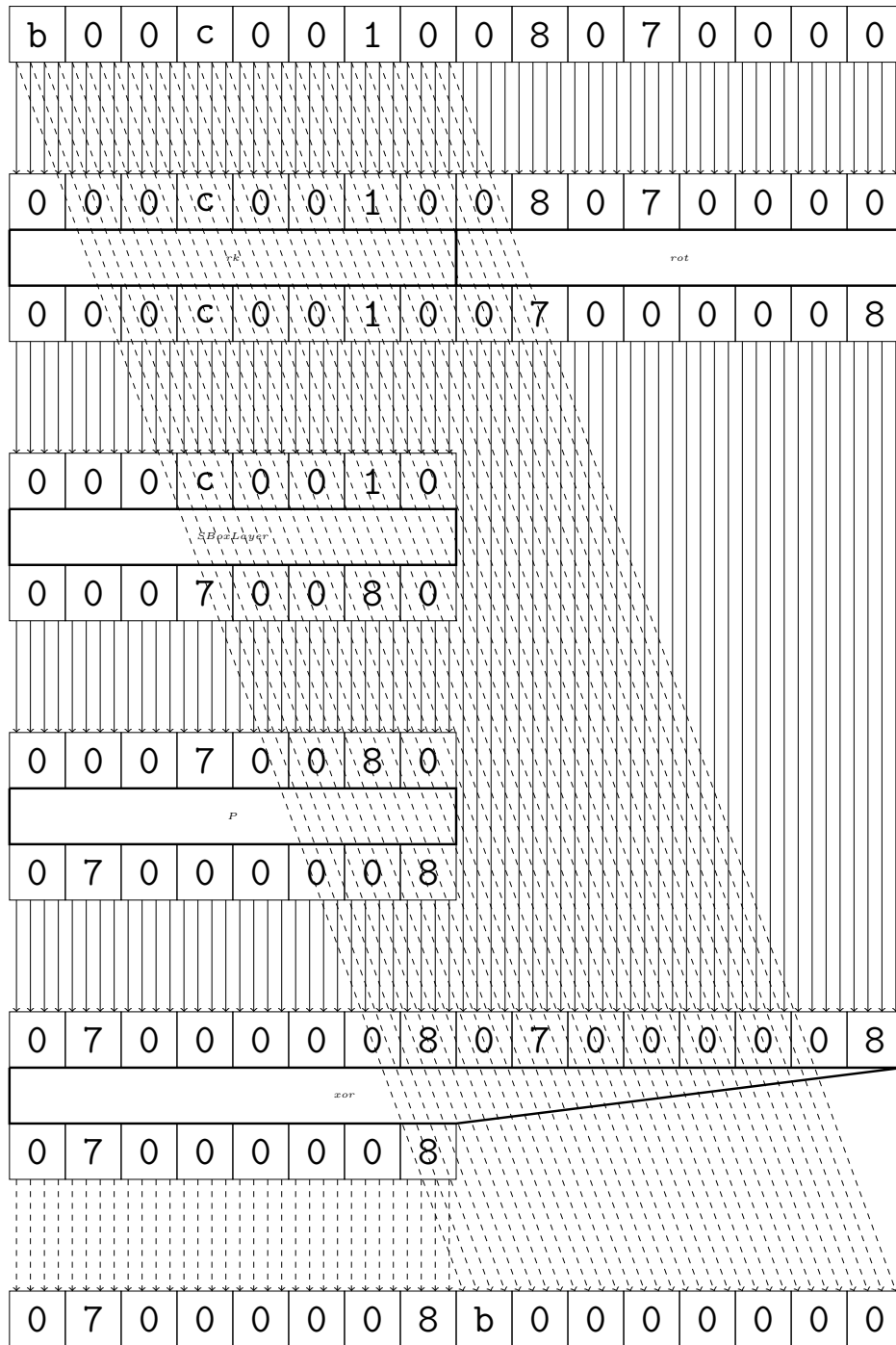


31 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

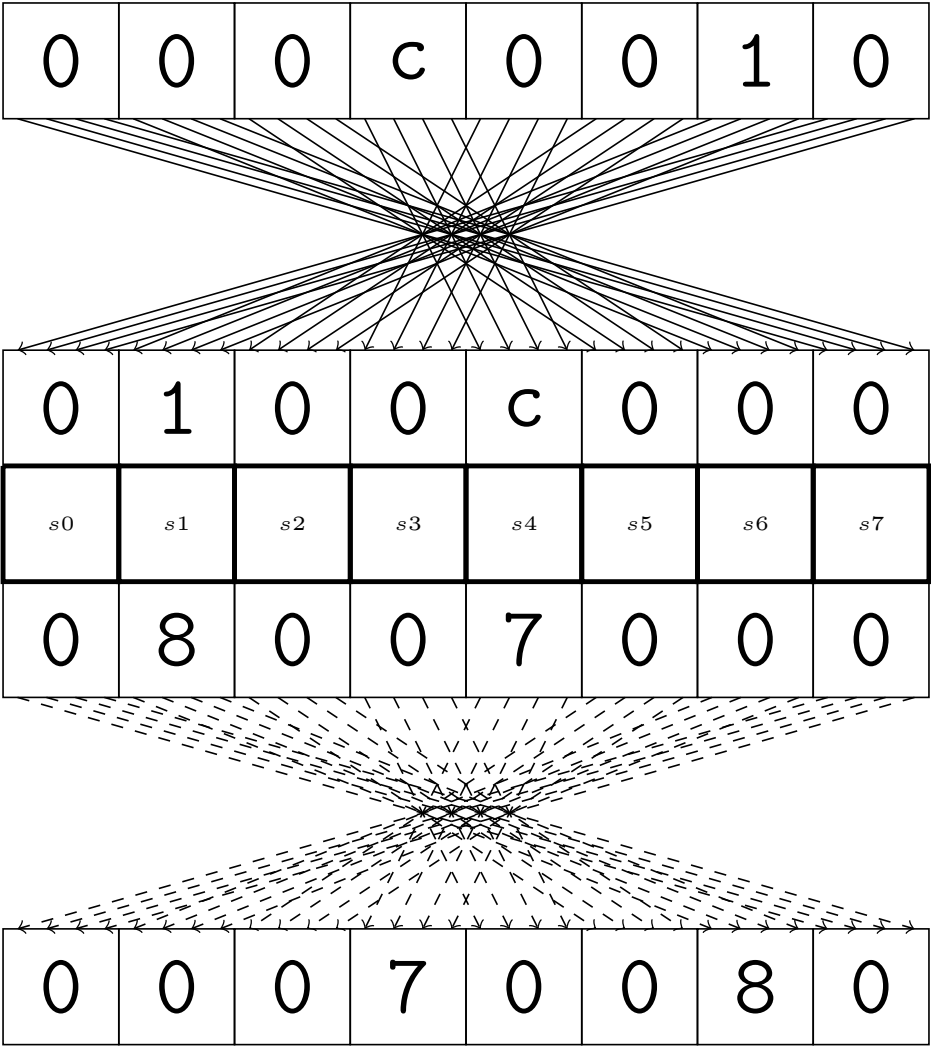


32 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

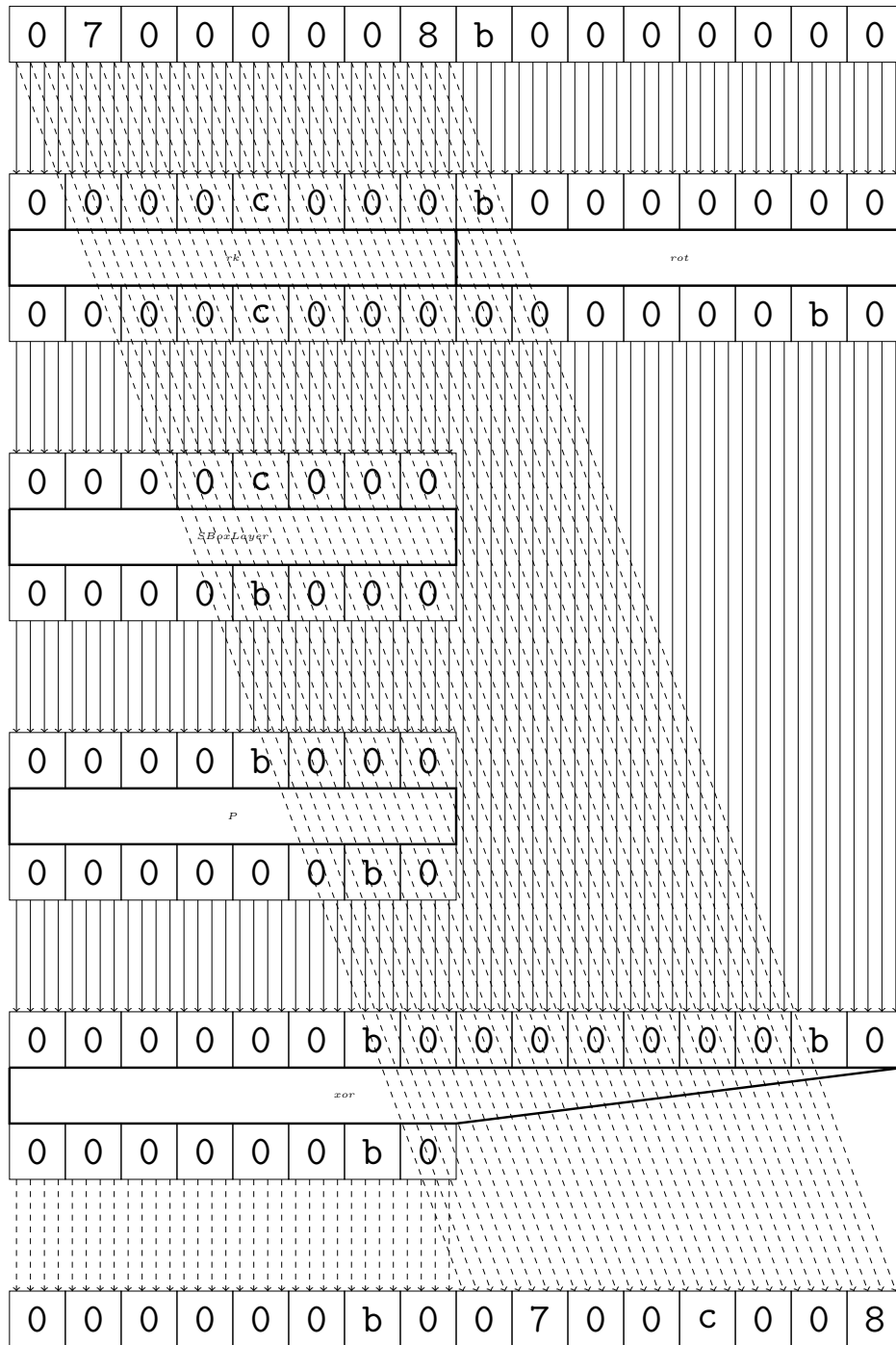
33 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



34 LBlock_round

Maximal linear correlation: $2^{-1.0}$



35 SBoxLayer

Maximal linear correlation: $2^{-1.0}$

